Homological Algebra

Labix

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Abstract

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1 Abelian Categories and Exact Sequences

1.1 Additive Categories

Definition 1.1.1 (Pre-Additive Categories)
Let $\mathcal{C}$ be a category. We say that $\mathcal{C}$ is pre-additive if the following is true.

- For any $X, Y \in \mathcal{C}$, the Hom set
$$\text{Hom}_{\mathcal{C}}(X, Y) \in \mathbf{Ab}$$
has the structure of an abelian group.
- For any $X, Y, Z \in \mathcal{C}$, the composition of morphisms
$$\circ : \text{Hom}_{\mathcal{C}}(X, Y) \times \text{Hom}_{\mathcal{C}}(Y, Z) \to \text{Hom}_{\mathcal{C}}(X, Z)$$
is bilinear. This means that if $f, g \in \text{Hom}_{\mathcal{C}}(X, Y)$ and $h, k \in \text{Hom}_{\mathcal{C}}(Y, Z)$ are morphisms, then
$$h \circ (f + g) = h \circ f + h \circ g \quad \text{and} \quad (h + k) \circ f = h \circ f + k \circ f$$

We also say that $\mathcal{C}$ is enriched over $\mathbf{Ab}$. This relates to enriched category theory.

Definition 1.1.2 (Additive Categories) A category $\mathcal{A}$ is additive if in addition to being pre-additive, it also satisfies the following:

- $\mathcal{A}$ has a zero object, denoted 0
- $\mathcal{A}$ has finite products

Lemma 1.1.3
Let $\mathcal{A}$ be an additive category. Let $X, Y \in \mathcal{A}$. Then the universal property of limits and colimits induces an isomorphism
$$X \times Y \cong X \coprod Y$$

Proof
I first claim that $X \times 0 \cong X$. Since 0 is a zero object, there exists a unique morphism $X \to 0$. Together with the identity map $X \to X$, the universal property of products imply that there exists a unique morphism $X \to X \times 0$ such that $X \to X \times 0 \to X$ is the identity. Precomposing this composite with the projection map $X \times 0 \to X$ gives $X \times 0 \to X \to X \times 0 \to X = X \times 0 \to X$. Since the identity map in a category is unique, we must have that $X \times 0 \to X \to X \times 0 = \text{id}_X$. Hence $X \times 0 \cong X$.

Since 0 is initial, there exists unique morphisms $(\text{id}_X, 0) : X \times 0 \to X \times Y$ and $(0, \text{id}_Y) : 0 \times Y \to X \times Y$ such that further composing with projection maps $\text{pr}_X : X \times Y \to X$ and $\text{pr}_Y : X \times Y \to 0 \times Y$ give the identity. I claim that these maps give $X \times Y$ the universal property of coproducts. Let $Z \in \mathcal{A}$ be an object and let $f_X : X \to Z$ and $f_Y : Y \to Z$ be morphisms. Define a morphism $X \times Y \to Z$ by $f_X \circ \text{pr}_X + f_Y \circ \text{pr}_Y$. We need to show that the following diagram is commutative:

$$\begin{tikzcd}
 & Z & \\
 & X \times Y \arrow[u, "f_X \circ \text{pr}_X + f_Y \circ \text{pr}_Y"] & \\
X \times 0 \arrow[uur, bend left, "f_X"] \arrow[ur, "(\text{id}_X, 0)"'] & & 0 \times Y \arrow[uul, bend right, "f_Y"'] \arrow[ul, "(0, \text{id}_Y)"]
\end{tikzcd}$$

Indeed, we have

$$\begin{aligned}(f_X \circ \text{pr}_X + f_Y \circ \text{pr}_Y) \circ (\text{id}_X, 0) &= f_X \circ \text{pr}_X \circ i_X + f_Y \circ \text{pr}_Y \circ 0 && (\mathcal{A} \text{ is pre-additive}) \\ &= f_X \circ \text{id}_X + f_Y \circ 0 \\ &= f_X + 0 \\ &= f_X\end{aligned}$$

and similarly we have $(f_X \circ \text{pr}_X + f_Y \circ \text{pr}_Y) \circ (\text{id}_Y, 0) = f_Y$.

It remains to show that this map is unique such that the compositions $X \times 0 \to X \times Y \to Z$ and $0 \times Y \to X \times Y \to Z$ are the maps $f_X : X \to Z$ and $f_Y : Y \to Z$ respectively. Suppose that ϕ is another such map. I first claim that

$$(\text{id}_X, 0) \circ \text{pr}_X + (0, \text{id}_Y) \circ \text{pr}_Y = \text{id}_{X \times Y}$$

We have that

$$\begin{aligned}\phi &= \phi \circ \text{id}_{X \times Y} \\ &=\end{aligned}$$

■

1.2 Abelian Categories

Definition 1.2.1 (Zero Morphism)
Let $\mathcal{A}$ be a category with a zero object. Let $C, D \in \mathcal{A}$ be objects. Define the zero morphism from C to D to be the the unique morphism

$$0 : C \overset{\exists!}{\to} 0 \overset{\exists!}{\to} D$$

Definition 1.2.2 (Kernels and Cokernels)
Let $\mathcal{A}$ be a category with a zero object. Let $C, D \in \mathcal{A}$ be objects. Let $f : C \to D$ be a morphism.

- Define the kernel of f to be
$$\ker(f) = \text{Eq}(f, 0)$$
if it exists.
- Define the cokernel of f to be
$$\text{coker}(f) = \text{Coeq}(f, 0)$$
if it exists.

Definition 1.2.3 (Pre-Abelian Category)
Let $\mathcal{A}$ be an additive category. We say that $\mathcal{A}$ is a pre-abelian category if every morphisms in $\mathcal{A}$ has a kernel and a cokernel.

Lemma 1.2.4
Let $\mathcal{A}$ be pre-abelian category. Let $f : C \to D$ be a morphism. Then f factors naturally and uniquely as

$$C \to \text{coker}(\ker(f) \to C) \to \ker(D \to \text{coker}(f)) \to D$$

Proof Since $\ker(f) \to C \to D$ is 0, we invoke the universal property of cokernels to obtain the following commutative diagram:

$$\begin{tikzcd} & & \text{coker}(\ker(f) \to C) \arrow[d, dashed, "\exists!"] & \\ \ker(f) \arrow[r] & C \arrow[r] \arrow[ur] & D \arrow[r] & \text{coker}(f) \end{tikzcd}$$

Now since $\mathcal{A}$ is an abelian category, any epimorphism h satisfies $a \circ h = 0$ implies $a = 0$. Since $C \to \text{coker}(\ker(f) \to C)$ is an epimorphism by 1.2.7 and $C \to \text{coker}(\ker(f) \to C) \to D \to \text{coker}(f) = 0$, we conclude that

$$\text{coker}(\ker(f) \to C) \to D \to \text{coker}(f) = 0$$

Then by the universal property of kernels, we obtain the following commutative diagram:

$$\begin{tikzcd}
\text{coker}(\ker(f) \to C) \arrow[d, dashed, "\exists!"'] \arrow[dr] & & \\
\ker(D \to \text{coker}(f)) \arrow[r] & D \arrow[r] & \text{coker}(f)
\end{tikzcd}$$

It is natural since all morphisms are induced by a universal property. ■

Definition 1.2.5 (Abelian Categories)
Let $\mathcal{A}$ be a pre-abelian category. We say that $\mathcal{A}$ is an abelian category if the following are true.
- Every monomorphism is the kernel of some morphism.
- Every epimorphism is the cokernel of some morphism.

Lemma 1.2.6
Let $\mathcal{A}$ be a pre-abelian category. Then the following are equivalent.
- $\mathcal{A}$ is an abelian category.
- Every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel.
- For all morphisms $f \in \mathcal{A}$, the unique morphism

$$\text{coker}(\ker(f) \to C) \xrightarrow{\cong} \ker(D \to \text{coker}(f))$$

  is an isomorphism.

Proof (2) $\implies$ (3): Since $\text{coker}(\ker(f) \to C) = \ker(D \to \text{coker}(f))$ we are done. ■

Example 1.2.7
Let R be a ring. Then ${}_R\mathbf{Mod}$ is an abelian category.

Proposition 1.2.8
Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C}$ be a small category. Then $\mathcal{A}^{\mathcal{C}}$ is an abelian category.

1.3 Injecltivity, Surjectivity and Images

Definition 1.3.1 (Injectivity and Surjectivity) Let $f : X \to Y$ be a morphism in an abelian category.
- We say that f is injective if $\ker(f) = 0$
- We say that f is surjective if $\text{coker}(f) = 0$

In particular, these notions coincide that of epics and monics in an abelian category.

Proposition 1.3.2 Let $f : X \to Y$ be a morphism in an abelian category. Then the following are true.
- f is injective if and only if f is a monomorphism
- f is surjective if and only if f is epimorphism

Lemma 1.3.3
Let $\mathcal{A}$ be an abelian category. Let $f : X \to Y$ be a morphism. Then the following are true.
- The morphism $\ker(f) \to X$ is injective.

- The morphism $Y \to \text{coker}(f)$ is surjective.

Proof Let $\iota : \ker(f) \to X$ be universal morphism. Let $a, b : Z \to \ker(f)$ be morphisms such that $\iota \circ a = \iota \circ b$. Applying the universal property of kernels. We know that $f \circ \iota \circ a = f \circ \iota \circ b = 0$ since $f \circ \iota = 0$. Therefore we can apply the universal property of kernels to $\iota \circ a = \iota \circ b$ show that there exists a unique morphism $u : Z \to \ker(f)$ such that $\iota \circ u = \iota \circ a = \iota \circ b$. But since u is unique, we must have $a = u = b$.

The proof for cokernels is dual. ■

Definition 1.3.4 (The Image of a Morphism)
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathcal{A}$. Let $f : C \to D$ be a morphism. Define the image of f to be

$$\text{im}(f) = \ker(D \to \text{coker}(f)) \cong \text{coker}(\ker(f) \to C)$$

Indeed, thinking in terms of modules over a ring R, the cokernel of an R-module homomorphism $f : M \to N$ is $\frac{N}{\text{im}(f)}$ together with the canonical map $N \to \frac{N}{\text{im}(f)}$. Then the kernel of this map is precisely $\text{im}(f)$. On the other hand, the cokernel of the morphism $ker(f) \to M$ is $\frac{M}{\ker(f)}$ since the inclusion map is injective, and by the first isomorphism theorem $\frac{M}{\ker(f)} \cong \text{im}(f)$.

Lemma 1.3.5
Let $\mathcal{A}$ be an abelian category. Let $A, B \in \mathcal{A}$ be objects. Let $f : A \to B$ be a morphism. Then there exists a natural and unique (up to isomorphism) factorization of f into an epimorphism followed by an monomorphism:

$$\begin{tikzcd} A \arrow[rr, "f"] \arrow[dr, two heads, dashed, "\exists!"'] & & B \\ & \exists! C \arrow[ur, tail, dashed, "\exists!"'] & \end{tikzcd}$$

Such an object that f factors through is isomorphic to $\text{im}(f)$.

Proof
We have seen that the map $C \to \text{coker}(\ker(f) \to C)$ is an epimorphism in lmm1.2.10. Similarly, we have seen that the map $\ker(D \to \text{coker}(f)) \to D$ is a monomorphism. The factorization is clearly natural. It remains to show that $\text{im}(f)$ is the unique object such that f factors through as an epimorphism followed by a monomorphism. ■

Lemma 1.3.6
Let $\mathcal{A}$ be an abelian category. Let $A, B \in \mathcal{A}$ be objects. Let $f : A \to B$ be a morphism. Then the following are true.
- If f is injective, then $A \cong \text{im}(f)$.
- If f is surjective, then $\text{im}(f) \cong B$.

Proof
We know that $A \to \text{im}(f)$ is surjective by the above. Since $f = A \to \text{im}(f) \to B$ and f is injective, we conclude that $A \to \text{im}(f)$ is injective.

Similarly, by the above lemma we know that $\text{im}(f) \to B$ is injective. Since $f = A \to \text{im}(f) \to B$ and f is surjective, we conclude that $\text{im}(f) \to B$ is surjective. ■

1.4 Additive Functors

Definition 1.4.1 (Additive Functors) Let $\mathcal{A}, \mathcal{B}$ be abelian categories. We say that a functor $F : \mathcal{A} \to \mathcal{B}$ is additive if for every $X, Y \in \mathcal{A}$, the map

$$\text{Hom}_{\mathcal{A}}(X, Y) \to \text{Hom}_{\mathcal{B}}(F(X), F(Y))$$

is a homomorphism of abelian groups.

Proposition 1.4.2
Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor. Then the following are equivalent.
- F is additive.
- F preserves the zero object and binary direct sums.
- F preserves finite products.
- F preserves finite coproducts.

1.5 Exact Sequences

Definition 1.5.1 (Exact Sequences)
Let $\mathcal{A}$ be an abelian category. Let the following be sequence of morphisms in $\mathcal{A}$:

$$A \xrightarrow{f} B \xrightarrow{g} C$$

We say that the sequence is exact at B if the following are true.
- $g \circ f = 0$.
- Since $g \circ f = 0$, we know that g factors through $\text{coker}(f)$. Then the composite $\text{im}(f) = \ker(B \to \text{coker}(f)) \to B \xrightarrow{g} C$ is 0 since $\text{im}(f) \to B \to \text{coker}(f) = 0$. By the universal property of kernels, there exists a unique morphism $\text{im}(f) \to \ker(g)$:

$$\begin{tikzcd}
\text{im}(f) = \ker(B \to \text{coker}(f)) \arrow[d, dashed, "\exists!"] \arrow[dr] & & \\
\ker(g) \arrow[r] & B \arrow[r] \arrow[dr, "g"'] & \text{coker}(f) \arrow[d, dashed, "\exists!"] \\
& & C
\end{tikzcd}$$

We require that this morphism is an isomorphism.

Definition 1.5.2 (Short Exact Sequence)
Let $\mathcal{A}$ be an abelian category. A short exact sequence in $\mathcal{A}$ is a diagram of the form

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

such that the sequence is exact in A, B and C.

Lemma 1.5.3
Let $\mathcal{A}$ be an abelian category. Then the following are true.
- Let $0 \to A \to B \to C$ be a sequence of morphisms in $\mathcal{A}$. Then it is exact if and only if $g \circ f = 0$ and $A \cong \ker(g)$.
- Let $A \to B \to C \to 0$ be a sequence of morphisms in $\mathcal{A}$. Then it is exact if and only if $g \circ f = 0$ and $C \cong \text{coker}(g)$.

Proof
Suppose that $g \circ f = 0$ and $A \cong \ker(g)$. We first show exactness at A. Trivially, we have $0 \to A \to B = 0$. Moreover, $\text{im}(0 \to A) = 0 = \ker(\ker(g) \to B)$. Now we show exactness at B. We have

$g \circ f = 0$ already by assumption, and

$$\operatorname{im}(\ker(g) \to B) = \operatorname{coker}(\ker(\ker(g) \to B) \to \ker(g)) = \operatorname{coker}(0 \to \ker(g)) = \ker(g)$$

Hence the sequence is exact.

Now suppose that $0 \to A \to B \to C$ is an exact sequence. Then clearly we have $g \circ f = 0$. To show that $A \cong \ker(g)$, we appeal to the universal property of kernels. Since $A \to B \to C = 0$, the morphism $f : A \to B$ must factor through $\ker(B \to C)$, hence we obtain the following diagram:

$$\begin{tikzcd}
A \arrow[rr, tail, "f"] \arrow[dr, two heads] \arrow[drr, tail] & & B \\
& \operatorname{im}(f) \arrow[r, tail, "\cong"'] & \ker(g) \arrow[u, tail]
\end{tikzcd}$$

Since $A \to \operatorname{im}(f) \to \ker(g)$ is surjective, we have that $A \to \ker(g)$ is surjective. Since f is injective, we have that $A \to \ker(g)$ is injective. Hence $A \cong \ker(g)$.

The second statement is entirely dual to the first statement and so we omit it. ■

Proposition 1.5.4 (Decomposing Long Exact Sequences into Short Exact Sequences)
Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, \partial_\bullet)$ be an exact sequence in $\mathcal{A}$. Then the diagonal sequences in the following diagram are short exact sequences:

$$\begin{tikzcd}
& & 0 \arrow[dr] & & 0 & & \\
\cdots \arrow[dr] & & & \operatorname{im}(\partial_{n+1}) \arrow[dr] \arrow[ur] & & & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[rr, "\partial_{n+2}"] \arrow[dr] & & C_{n+1} \arrow[rr, "\partial_{n+1}"] \arrow[ur] & & C_n \arrow[rr, "\partial_n"] \arrow[dr] & & C_{n-1} \arrow[r] \arrow[ur] & \cdots \\
& & \operatorname{im}(\partial_{n+2}) \arrow[ur] \arrow[dr] & & & & \operatorname{im}(\partial_n) \arrow[ur] \arrow[dr] & & \\
& 0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0 &
\end{tikzcd}$$

Proof Suppose we are given the long exact sequence and consider the 3 term sequence below. It is clear that the inclusion map is injective. Similarly, the map $f_i : M_i \to \operatorname{im}(f_i)$ is surjective tautologically. Finally, $\ker(f_i) = \operatorname{im}(f_{i+1})$ from the long exact sequence. ■

Proposition 1.5.5 (Gluing Short Exact Sequences into Long Exact Sequences)
Let $\mathcal{A}$ be an abelian category. Suppose that the following is a diagram in $\mathcal{A}$:

$$\begin{tikzcd}
& & 0 \arrow[dr] & & 0 & & \\
\cdots \arrow[dr] & & & D_n \arrow[dr, "f_n"] \arrow[ur] & & & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[rr, "f_{n+1} \circ g_{n+2}"] \arrow[dr, "g_{n+2}"'] & & C_{n+1} \arrow[rr, "f_n \circ g_{n+1}"] \arrow[ur, "g_{n+1}"] & & C_n \arrow[rr, "f_{n-1} \circ g_n"] \arrow[dr, "g_n"'] & & C_{n-1} \arrow[r] \arrow[ur] & \cdots \\
& & D_{n+1} \arrow[ur, "f_{n+1}"'] \arrow[dr] & & & & D_{n-1} \arrow[ur, "f_{n-1}"'] \arrow[dr] & & \\
& 0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0 &
\end{tikzcd}$$

such that all diagonal short exact sequences are exact. Then the horizontal sequence is

exact.

Proof Since h_i is surjective, we have that $\operatorname{im}(g_i \circ h_{i+1}) = \operatorname{im}(g_i)$. Similarly, since g_i is injective, we have that $\ker(g_{i-1} \circ h_i) = \ker(h_i)$. But from the short exact sequence we know that $\operatorname{im}(g_i) = \ker(h_i)$. Hence we have

$$\operatorname{im}(g_i \circ h_{i+1}) = \operatorname{im}(g_i) = \ker(h_i) = \ker(g_{i-1} \circ h_i)$$

so that we have a long exact sequence. ■

1.6 Properties of Exact Sequences

Theorem 1.6.1 (The Snake Lemma)
Let $\mathcal{A}$ be an abelian category. Let the following be a commutative diagram in $\mathcal{A}$:

$$\begin{tikzcd}
 & A \arrow[r, "f"] \arrow[d, "a"] & B \arrow[r, "g"] \arrow[d, "b"] & C \arrow[r] \arrow[d, "c"] & 0 \\
0 \arrow[r] & A' \arrow[r, "f'"] & B' \arrow[r, "g'"] & C' &
\end{tikzcd}$$

If the two rows are exact, then there exists a connecting homomorphism $\delta : \ker(c) \to \operatorname{coker}(a)$ such that the following sequence is exact:

$$\begin{tikzcd}
0 \arrow[r] & \ker(f) \arrow[r] & \ker(a) \arrow[r] & \ker(b) \arrow[r] & \ker(c) \arrow[dll, rounded corners] & & \\
 & & \operatorname{coker}(a) \arrow[r] & \operatorname{coker}(b) \arrow[r] & \operatorname{coker}(c) \arrow[r] & \operatorname{coker}(g') \arrow[r] & 0
\end{tikzcd}$$

Proof
We first show that the first three terms in the sequence are exact. ■

It is called the snake lemma because of the following diagram:

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & & \\
0 \arrow[r] & \ker(f) \arrow[r] & \ker(a) \arrow[r] \arrow[d] & \ker(b) \arrow[r] \arrow[d] & \ker(c) \arrow[d] \arrow[dll, rounded corners, "\delta"'] & & \\
 & & A \arrow[r, "f"] \arrow[d, "a"] & B \arrow[r, "g"] \arrow[d, "b"] & C \arrow[r] \arrow[d, "c"] & 0 & \\
 & 0 \arrow[r] & A' \arrow[r, "f'"'] \arrow[d] & B' \arrow[r, "g'"'] \arrow[d] & C' \arrow[d] & & \\
 & & \operatorname{coker}(a) \arrow[r] \arrow[d] & \operatorname{coker}(b) \arrow[r] \arrow[d] & \operatorname{coker}(c) \arrow[r] \arrow[d] & \operatorname{coker}(g') \arrow[r] & 0 \\
 & & 0 & 0 & 0 & &
\end{tikzcd}$$

where all rows and columns are exact, and δ connects the two exact sequences into another exact sequence. If objects of the abelian category $\mathcal{A}$ have underlying sets, we may describe δ explicitly via the formula

$$\delta(c) = ((f')^{-1} \circ b \circ g^{-1})(c)$$

Lemma 1.6.2 (Naturality of the Snake Lemma)
Let $\mathcal{A}$ be an abelian category. Let the following be a commutative diagram in $\mathcal{A}$:

$$\begin{tikzcd}
 & & D \arrow[rr] \arrow[dd, "d"] & & E \arrow[rr] \arrow[dd, "e"] & & F \arrow[rr] \arrow[dd, "f"] & & 0 \\
 & A \arrow[ur] \arrow[rr] \arrow[dd, "a"] & & B \arrow[ur] \arrow[rr] \arrow[dd, "b"] & & C \arrow[ur] \arrow[rr] \arrow[dd, "c"] & & 0 \arrow[ur] & \\
0 \arrow[rr] & & D' \arrow[rr] & & E' \arrow[rr] & & F' & & \\
0 \arrow[ur] \arrow[rr] & & A' \arrow[ur] \arrow[rr] & & B' \arrow[ur] \arrow[rr] & & C' \arrow[ur] & &
\end{tikzcd}$$

If the four rows are exact, then the above diagram induces the following commutative diagram:

$$\begin{tikzcd}
\ker(c) \arrow[r, "\delta"] \arrow[d] & \operatorname{coker}(a) \arrow[d] \\
\ker(f) \arrow[r, "\delta'"] & \operatorname{coker}(d)
\end{tikzcd}$$

where δ and δ' are the connecting homomorphisms given by the snake lemma.

Lemma 1.6.3 (Short Five Lemma)
Let $\mathcal{A}$ be an abelian category. Let the following be a commutative diagram in $\mathcal{A}$:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "f"] & B \arrow[r] \arrow[d, "g"] & C \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & 0
\end{tikzcd}$$

If any two of f, g, h are isomorphisms, then the third one is also an isomorphism.

Proof
Applying the snake lemma gives an exact sequence of the form

$$\ker(f) \to \ker(g) \to \ker(h) \to \operatorname{coker}(f) \to \operatorname{coker}(g) \to \operatorname{coker}(h)$$

If f and g are isomorphisms, then the snake lemma shows that $\ker(h) = 0$. Hence h is injective. Since $B \to B' \to C'$ is surjective, we have that $B \to C \to C'$ is surjective by commutativity of the diagram, hence $h : C \to C'$ is surjective. Thus h is an isomorphism.

If g and h are isomorphisms, then the snake lemma shows that $\operatorname{coker}(f) = 0$. Hence f is surjective. Since $A \to B \to B'$ is injective, we have that $A \to A' \to B'$ is also injective by commutativity of the diagram, hence $f : A \to A'$ is injective. Hence f is an isomorphism.

If f and h are isomorphisms, then the snake lemma shows that $\ker(g) = \operatorname{coker}(g) = 0$. Hence g is an isomorphism. ■

Theorem 1.6.4 (Five Lemma)
Let $\mathcal{A}$ be an abelian category. Let the following be a commutative diagram in $\mathcal{A}$:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "l"'] & B \arrow[r, "g"] \arrow[d, "m"'] & C \arrow[r, "h"] \arrow[d, "n"'] & D \arrow[r, "k"] \arrow[d, "p"] & E \arrow[d, "q"] \\
A' \arrow[r, "f'"'] & B' \arrow[r, "g'"'] & C' \arrow[r, "h'"'] & D' \arrow[r, "k'"'] & E'
\end{tikzcd}$$

Suppose that the following are true.
- The two rows are exact.
- $m : B \to B'$ and $p : D \to D'$ are isomorphisms.

- $l : A \to A'$ is an epimorphism.
- $q : E \to E'$ is a monomorphism.

Then $n : C \to C'$ is an isomorphism.

Proof
Using the factorization of a morphism through its image, we obtain the following commutative diagram:

$$\begin{tikzcd}
A \arrow[rr, "f"] \arrow[dr] \arrow[dd, "l"'] & & B \arrow[rr, "g"] \arrow[dr] \arrow[dd, "m"] & & C \arrow[rr, "h"] \arrow[dr] \arrow[dd, "n"] & & D \arrow[rr, "k"] \arrow[dr] \arrow[dd, "p"] & & E \arrow[dd, "q"] \\
& \mathrm{im}(f) \arrow[ur] \arrow[dd] & & \mathrm{im}(g) \arrow[ur] \arrow[dd] & & \mathrm{im}(h) \arrow[ur] \arrow[dd] & & \mathrm{im}(k) \arrow[ur] \arrow[dd] & \\
A' \arrow[rr, "f'"] \arrow[dr] & & B' \arrow[rr, "g'"] \arrow[dr] & & C' \arrow[rr, "h'"] \arrow[dr] & & D' \arrow[rr, "k'"] \arrow[dr] & & E' \\
& \mathrm{im}(f') \arrow[ur] & & \mathrm{im}(g') \arrow[ur] & & \mathrm{im}(h') \arrow[ur] & & \mathrm{im}(k') \arrow[ur] &
\end{tikzcd}$$

Since $A' \to \mathrm{im}(f')$ is surjective and $A \to A'$ is surjective, we have that $A \to \mathrm{im}(f) \to \mathrm{im}(f')$ is surjective. Hence $\mathrm{im}(f) \to \mathrm{im}(f')$ is surjective. On the other hand, Since $\mathrm{im}(f) \to B$ is injective and $B \to B'$ is an isomorphism, we have that $\mathrm{im}(f) \to \mathrm{im}(f') \to B'$ is injective, hence $\mathrm{im}(f) \to \mathrm{im}(f')$ is injective. Thus $\mathrm{im}(f) \to \mathrm{im}(f')$ is an isomorphism. Now since $\mathrm{im}(f) = \ker(g)$ and $\mathrm{im}(f') = \ker(g')$, we have that the rows in the following diagram are exact:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \mathrm{im}(f) \arrow[r] \arrow[d] & B \arrow[r] \arrow[d, "m"] & \mathrm{im}(g) \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \mathrm{im}(f') \arrow[r] & B' \arrow[r] & \mathrm{im}(g') \arrow[r] & 0
\end{tikzcd}$$

By the short five lemma and the fact that $m : B \to B'$ is an isomorphism, we conclude that $\mathrm{im}(g) \to \mathrm{im}(g')$ is an isomorphism. By an entirely dual argument starting from the right, we also conclude that $\mathrm{im}(h) \to \mathrm{im}(h')$ is an isomorphism. Finally, since $\mathrm{im}(g) = \ker(h)$ and $\mathrm{im}(g') = \ker(h')$, we have that the rows in the following diagram are exact:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \mathrm{im}(g) \arrow[r] \arrow[d] & C \arrow[r] \arrow[d, "n"] & \mathrm{im}(h) \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \mathrm{im}(g') \arrow[r] & C' \arrow[r] & \mathrm{im}(h') \arrow[r] & 0
\end{tikzcd}$$

Then by the short five lemma, we conclude that $n : C \to C'$ is an isomorphism. ■

Theorem 1.6.5 (The Nine Lemma)

Definition 1.6.6 (Split Complexes)
Let $\mathcal{A}$ be an abelian category. Let $C \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. We say that C is split if there exists maps $s_n : C_n \to C_{n+1}$ such that

$$\partial = \partial \circ s \circ \partial$$

If C is also exact, then we call C split exact.

Theorem 1.6.7 (The Splitting Lemma) Let $\mathcal{A}$ be an abelian category. Let $A, B, C \in \mathcal{A}$. Then the following are equivalent for a short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

- The short exact sequence is split exact.
- There is an isomorphism $B \cong A \oplus C$.

- There exists a morphism $p : B \to A$ such that $p \circ f = \text{id}_A$.
- There exists a morphism $s : C \to B$ such that $g \circ s = \text{id}_C$.

Proposition 1.6.8 Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. Then F preserves split exact sequences.

1.7 Exact Functors and the Freyd-Mitchell Embedding Theorem

Definition 1.7.1 (Exact Functors) Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. Let $0 \to A \to B \to C \to 0$ be an exact sequence in $\mathcal{A}$. We say that F is exact if the sequence

$$0 \to F(A) \to F(B) \to F(C) \to 0$$

is exact.

Example 1.7.2
Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Then the functor

$$S^{-1} : {}_R\mathbf{Mod} \to {}_{S^{-1}}R\mathbf{Mod}$$

is exact.

Lemma 1.7.3 Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor. If F is an equivalence of categories, then F is an exact functor.

Definition 1.7.4 (Abelian Subcategories)
Let $\mathcal{A}$ be an abelian category. Let $\mathcal{B}$ be a subcategory of $\mathcal{A}$. We say that $\mathcal{B}$ is an abelian subcategory of $\mathcal{B}$ if the following are true.
- $\mathcal{B}$ is an abelian category.
- The inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$ is an exact functor.

Example 1.7.5
Let R be a ring. Then ${}_R\mathbf{FGMod}$ is an abelian subcategory of ${}_R\mathbf{Mod}$ if and only if R is Noetherian.

The notion of abelian subcategories is rather subtle: since exact sequences in $\mathcal{B}$ must be exact in $\mathcal{A}$, this means that for example, kernels formed in $\mathcal{B}$ must coincide with the kernels formed in $\mathcal{A}$ with the morphism. This is not true in for example: the category of sheaves and the category of presheaves (they both are abelian categories but the category of sheaves is not an abelian subcategory of the category of presheaves)

Theorem 1.7.6 (Freyd-Mitchell Embedding Theorem)
Let $\mathcal{A}$ be a small abelian category. Then there exists a ring R and an exact, fully faithful embedding $F : \mathcal{A} \to {}_R\mathbf{Mod}$.

Such a functor allows us think about diagrams in $\mathcal{A}$ as if they were diagrams in ${}_R\mathbf{Mod}$

1.8 Adjunctions in the Category of Modules

Theorem 1.8.1 (Tensor-Hom Adjunction) Let R be a ring and let M be an R-module. Then there is an adjunction

$$- \otimes_R M : {}_R\mathbf{Mod} \rightleftarrows {}_R\mathbf{Mod} : \text{Hom}(M, -)$$

Explicitly, there is a natural isomorphism

$$\text{Hom}_{{}_R\mathbf{Mod}}(- \otimes_R M, -) \cong \text{Hom}_{{}_R\mathbf{Mod}}(-, \text{Hom}_{{}_R\mathbf{Mod}}(M, -))$$

More generally, the tensor-hom adjunction is a phenomena that exhibits ${}_R\mathbf{Mod}$ as a closed monoidal

category.

Proposition 1.8.2 Let R be a ring. Then $({}_R\mathbf{Mod}, \otimes_R, R)$ is a symmetric monoidal category. Moreover, it is a closed monoidal category with internal hom given by the usual hom functor $\text{HOM}_{{}_R\mathbf{Mod}}(-,-) = \text{Hom}_R(-,-)$.

We have seen that the left and right Hom functors

$$\text{Hom}(M,-) : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod} \quad \text{and} \quad \text{Hom}(-,M) : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod}$$

are left exact functors. Using the tensor-hom adjunction, we can deduce the following.

Proposition 1.8.3 Let R be a ring and let M be an R-module. Then both

$$- \otimes_R M : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod} \quad \text{and} \quad M \otimes_R - : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod}$$

are right exact functors.

We find it important to notice the relation between all four functor here. Notice that because there is a canonical isomorphism $M \otimes_R N \cong N \otimes_R M$, we can replace $- \otimes_R M$ in the natural isomorphism

$$\text{Hom}_{{}_R\mathbf{Mod}}(- \otimes_R M, -) \cong \text{Hom}_{{}_R\mathbf{Mod}}(-, \text{Hom}_{{}_R\mathbf{Mod}}(M,-))$$

with $M \otimes_R -$. This leaves us with the question: What is the functor $\text{Hom}(-,M)$ adjoint to? The answer is itself!

$$\text{Hom}_R(A, \text{Hom}_R(B,C)) \cong \text{Hom}(B, \text{Hom}(A,C))$$

Therefore it is more often to discuss $\text{Hom}(M,-)$ in conjunction with $- \otimes_R M$, while $\text{Hom}(-,M)$ is often left behind.

Theorem 1.8.4 (Change of Base Ring I)
Let R, S be commutative rings. Let $\varphi : R \to S$ be a ring homomorphism. Then there is an adjunction

$$- \otimes_R S : \mathbf{Mod}_R \overset{\dashv}{\rightleftarrows} \mathbf{Mod}_S : -|_R$$

Theorem 1.8.5 (Change of Base Ring II)
Let R, S be commutative rings. Let $\varphi : R \to S$ be a ring homomorphism. Then there is an adjunction

$$-|_R : \mathbf{Mod}_S \overset{\dashv}{\rightleftarrows} \mathbf{Mod}_R : \text{Hom}_R(S,-)$$

1.9 Graded Objects and Filtrations

Definition 1.9.1 (Graded Objects)
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. A G-graded object in $\mathcal{A}$ is a collection

$$C_\bullet = (C_g)_{g \in G}$$

for $C_g \in \mathcal{A}$.

Definition 1.9.2 (Morphisms of Graded Objects)
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Let $C_\bullet, D_\bullet$ be G-graded objects. A morphism from $C_\bullet$ to $D_\bullet$ is a collection

$$f_\bullet = (f_g)_{g \in G}$$

for $f_g : C_g \to D_g \in \mathcal{A}$.

Definition 1.9.3 (The Category of Graded Objects)
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Define the category

$$\mathbf{Gr}_G(\mathcal{A})$$

of G-graded objects as follows.
- Objects of $\mathbf{Gr}_G(\mathcal{A})$ are graded objects of $\mathcal{A}$.
- Morphisms of $\mathbf{Gr}_G(\mathcal{A})$ are the morphisms of graded objects.
- Composition is given by index-wise composition.

Lemma 1.9.4
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Then $\mathbf{Gr}_G(\mathcal{A})$ is an abelian category.

Definition 1.9.5 (The Shifting Functor)
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Let $g_0 \in G$. Define the gth shifting functor

$$-[g_0] : \mathbf{Gr}_G(\mathcal{A}) \to \mathbf{Gr}_G(\mathcal{A})$$

as follows.
- For each object $C_\bullet$, define $C_\bullet[g_0] = (C_{g+g_0})_{g\in G}$.
- For each morphisms $f_\bullet : C_\bullet \to D_\bullet$, define $f_\bullet[g] = (f_{g+g_0})_{g\in G}$.

Definition 1.9.6 (Morphisms of Given Degree)
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Let $g_0 \in G$. Let $C_\bullet, D_\bullet \in \mathbf{Gr}_G(\mathcal{A})$. A morphism from $C_\bullet$ to $D_\bullet$ of degree g_0 is a morphism $C_\bullet \to D_\bullet[g_0]$.

Lemma 1.9.7
Let G be an abelian group. Let $\mathcal{A}$ be an abelian category. Let $g_0 \in G$. Then there is a natural isomorphism

$$\text{Hom}_{\mathbf{Gr}_G(\mathcal{A})}(C_\bullet, D_\bullet[g_0]) \cong \text{Hom}_{\mathbf{Gr}_G(\mathcal{A})}(C_\bullet[-g_0], D_\bullet)$$

given by $f \mapsto f[-g_0]$.

Definition 1.9.8 (Filtration of an Object)
Let $\mathcal{A}$ be an abelian category. Let $A \in \mathcal{A}$.
- A descending filtration of A is a collection $\mathcal{F} = (F_nA)_{n\geq n_0}$ such that there are monomorphisms $F_{n+1}A \to F_nA$. We denote the filtration by

$$0 \subseteq \cdots \subseteq F_{n+1}A \subseteq F_nA \subseteq \cdots \subseteq A$$

- An ascending filtration of A is a collection $\mathcal{F} = (F_nA)_{n\geq n_0}$ such that there are monomorphisms $F_nA \to F_{n+1}A$. We denote the filtration by

$$0 \subseteq \cdots \subseteq F_nA \subseteq F_{n+1}A \subseteq \cdots \subseteq A$$

Definition 1.9.9 (Types of Filtrations)
Let $\mathcal{A}$ be an abelian category. Let $A \in \mathcal{A}$. Let $(F_nA)_{n\in\mathbb{N}^+}$ be a filtration of A.
- We say that the filtration is finite if there exists $s \in \mathbb{N}^+$ such that $F_sA = 0$.
- We say that the filtration is exhaustive if $\text{colim}_{n\to\infty} F_nA \cong A$.
- We say that the filtration if Hausdorff if $\lim_{n\to\infty} F_nA \cong 0$.

2 Chain Complexes

2.1 The Category of Chain and Cochain Complexes

Definition 2.1.1 (Chain Complexes)
Let $\mathcal{A}$ be an abelian category. A chain complex $(C_\bullet, \partial_\bullet^C)$ in $\mathcal{A}$ consists of the following data:
- A family of objects $C_n \in \mathcal{A}$ for $n \in \mathbb{Z}$
- Morphisms $\partial_n^C : C_n \to C_{n-1}$ in $\mathcal{A}$ for $n \in \mathbb{Z}$.

such that the chain condition:
$$\partial_n \circ \partial_{n+1} = 0$$
is satisfied for all $n \in \mathbb{Z}$.

We often write a chain complex in a diagram:
$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial_{n+1}^C} C_n \xrightarrow{\partial_n^C} C_{n-1} \longrightarrow \cdots$$
It is also common to denote a chain complex as C with the differential being implicit.

A chain complex is a $\mathbb{Z}$-graded object in an abelian category together with a structure map $d : C_\bullet \to C_\bullet[-1]$ satisfying $d \circ d = 0$.

Definition 2.1.2 (Chain Map)
Let $\mathcal{A}$ be an abelian category. Let C, D be two chain complexes in $\mathcal{A}$. A chain map $f : C \to D$ is a family of maps
$$f_n : C_n \to D_n$$
in $\mathcal{A}$ for $n \in \mathbb{Z}$ such that the chain map condition:
$$\partial_n^D \circ f_n = f_{n-1} \circ \partial_n^C$$
is satisfied for all $n \in \mathbb{Z}$.

In other words, we have the following commutative diagram:
$$\begin{tikzcd}
\cdots \arrow[r] & C_{n+1} \arrow[r, "\partial_{n+1}^C"] \arrow[d, "f_{n+1}"'] & C_n \arrow[r, "\partial_n^C"] \arrow[d, "f_n"'] & C_{n-1} \arrow[r] \arrow[d, "f_{n-1}"] & \cdots \\
\cdots \arrow[r] & D_{n+1} \arrow[r, "\partial_{n+1}^D"'] & D_n \arrow[r, "\partial_n^D"'] & D_{n-1} \arrow[r] & \cdots
\end{tikzcd}$$

Lemma 2.1.3
Let $\mathcal{A}$ be an abelian category. The following are true regarding chain maps.
- Let C, D, E be chain complexes in $\mathcal{A}$. Let $f : C \to D$ and $g : D \to E$ be chain maps. Then $g \circ f$ defined by $(g \circ f)_n = g_n \circ f_n$ is also a chain map.
- Let C be a chain complex in $\mathcal{A}$. Then $\mathrm{id} : C \to C$ defined by $\mathrm{id}_n = \mathrm{id}_{C_n}$ is also a chain map.

Proof
Since f and g are chain maps, we have that $\partial_n^D \circ f_n = f_{n-1} \circ \partial_n^C$ and $\partial_n^E \circ g_n = g_{n-1} \circ \partial_n^D$ for all $n \in \mathbb{Z}$. Then we have
$$\partial_n^E \circ g_n \circ f_n = g_{n-1} \circ \partial_n^D \circ f_n = g_{n-1} \circ f_{n-1} \circ \partial_n^C$$
for all $n \in \mathbb{Z}$ and so we are done.

The condition on chain maps is trivial with the identity map. ■

Definition 2.1.4 (Category of Chain Complexes)
Let $\mathcal{A}$ be an abelian category. Define
$$\mathbf{Ch}(\mathcal{A})$$

to be the category of chain complexes where
- The objects are chain complexes of objects in $\mathcal{A}$.
- The morphisms are chain maps.
- Composition is given by composition of functions pointwise.

Definition 2.1.5 (Bounded Chain Complexes)
Let $\mathcal{A}$ be an abelian category. Let $C \in \mathbf{Ch}(\mathcal{A})$ be a chain complex.
- We say that C is bounded above if there exists $N \in \mathbb{Z}$ such that $C_n = 0$ for all $n > N$.
- We say that C is bounded below if there exists $N \in \mathbb{Z}$ such that $C_n = 0$ for all $n < N$.
- We say that C is bounded if it is bounded above and bounded below.

Definition 2.1.6 (Variants of the Category of Chain Complexes)
Let $\mathcal{A}$ be an abelian category.
- Define the category
$$\mathbf{Ch}_{\geq 0}(\mathcal{A})$$
of non-negative chain complexes to be the full subcategory of $\mathbf{Ch}(\mathcal{A})$ consisting of chain complexes that is 0 in negative degrees.
- Define the category
$$\mathbf{Ch}_{\leq 0}(\mathcal{A})$$
of non-positive chain complexes to be the full subcategory of $\mathbf{Ch}(\mathcal{A})$ consisting of chain complexes that is 0 in positive degrees.

Proposition 2.1.7
Let $\mathcal{A}$ be an abelian category. Then $\mathbf{Ch}(\mathcal{A})$ is also an abelian category.

Thus $\mathbf{Ch}(\mathbf{Ch}(\mathcal{A}))$ also makes sense as a category. The objects are given by chain complexes of chain complexes.

Definition 2.1.8 (Cochain Complexes)
Let $\mathcal{A}$ be an abelian category. A cochain complex $(C^\bullet, \partial_C^\bullet)$ in $\mathcal{A}$ consists of the following data:
- A family of objects $C^n \in \mathcal{A}$ for $n \in \mathbb{Z}$
- Morphisms $\partial_C^n : C^n \to C^{n+1}$ in $\mathcal{A}$ for $n \in \mathbb{Z}$.

such that the chain condition:
$$\partial^n \circ \partial^{n-1} = 0$$
is satisfied for all $n \in \mathbb{Z}$.

We often write a cochain complex in a diagram:
$$\cdots \longrightarrow C^{n-1} \xrightarrow{\partial_C^{n-1}} C^n \xrightarrow{\partial_C^n} C^{n+1} \longrightarrow \cdots$$
Notice that when we write a cochain complex in diagram form, the arrows go the same way but the indexing increases in the opposite direction.

Algebraically speaking, there is no difference between a chain complex and a cochain complex, other than the fact that the boundary maps run in the other direction. For $(C^\bullet, \partial^\bullet)$ a cochain complex, we can form a chain complex by setting $C_n = C^{-n}$ and then using the same boundary maps.

Definition 2.1.9 (Chain Maps)
Let $\mathcal{A}$ be an abelian category. Let C, D be two chain complexes in $\mathcal{A}$. A chain map $f : C \to D$ is a family of maps
$$f^n : C^n \to D^n$$
in $\mathcal{A}$ for $n \in \mathbb{Z}$ such that the chain map condition:
$$\partial_D^n \circ f_n = f_{n-1} \circ \partial_C^n$$
is satisfied for all $n \in \mathbb{Z}$.

In other words, we have the following commutative diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "\partial_C^{n-1}"] \arrow[d, "f^{n-1}"'] & C^n \arrow[r, "\partial_C^n"] \arrow[d, "f^n"'] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] & \cdots \\
\cdots \arrow[r] & D^{n-1} \arrow[r, "\partial_D^{n-1}"'] & D^n \arrow[r, "\partial_D^n"'] & D^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Definition 2.1.10 (Category of Cochain Complexes)
Let $\mathcal{A}$ be an abelian category. Define the category of cochain complexes

$$\mathbf{CCh}(\mathcal{A})$$

to consist of the following data.
- The objects are the cochain complexes over $\mathcal{A}$
- The morphisms are the chain maps
- Composition is given by the composition of maps on all levels of the chain map.

Definition 2.1.11 (Bounded Cochain Complexes)
Let $\mathcal{A}$ be an abelian category. Let $C \in \mathbf{CCh}(\mathcal{A})$ be a cochain complex.
- We say that C is bounded above if there exists $N \in \mathbb{Z}$ such that $C_n = 0$ for all $n > N$.
- We say that C is bounded below if there exists $N \in \mathbb{Z}$ such that $C_n = 0$ for all $n < N$.
- We say that C is bounded if it is bounded above and bounded below.

Definition 2.1.12 (Variants of the Category of Cochain Complexes)
Let $\mathcal{A}$ be an abelian category.
- Define the category
$$\mathbf{CCh}_{\geq 0}(\mathcal{A})$$
of non-negative cochain complexes to be the full subcategory of $\mathbf{CCh}(\mathcal{A})$ consisting of cochain complexes that is 0 in negative degrees.
- Define the category
$$\mathbf{CCh}_{\leq 0}(\mathcal{A})$$
of non-positive chain complexes to be the full subcategory of $\mathbf{CCh}(\mathcal{A})$ consisting of cochain complexes that is 0 in positive degrees.

Definition 2.1.13 (The Inclusion Functors)
Let $\mathcal{A}$ be an abelian category. Let $n \in \mathbb{Z}$. Define the inclusion functor

$$c_n : \mathcal{A} \to \mathbf{Ch}(\mathcal{A})$$

by the following.
- For each $X \in \mathcal{A}$, we have $c_n(X)$ is the chain complex whose nth position is X and 0 for all the others, together with the 0 maps.
- For each morphism $f : X \to Y$, define $c_n(f) : c_n(X) \to c_n(Y)$ to be the chain map whose nth position is f and 0 in all other others.

Define the inclusion functor
$$c^n : \mathcal{A} \to \mathbf{CCh}(\mathcal{A})$$
by the following.
- For each $X \in \mathcal{A}$, we have $c^n(X)$ is the cochain complex whose nth position is X and 0 for all the others, together with the 0 maps.
- For each morphism $f : X \to Y$, define $c^n(f) : c^n(X) \to c^n(Y)$ to be the cochain map whose nth position is f and 0 in all other others.

The inclusion functors are in general not fully faithful.

2.2 Homology and Cohomology

Definition 2.2.1 (The Homology Groups of a Chain Complex)
Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, \partial_\bullet)$ be a chain complex in $\mathcal{A}$.

- Define the n-cycles of the chain complex to be

$$Z_n(C_\bullet) = \ker(\partial_n)$$

- Define the n-boundaries of the chain complex to be

$$B_n(C_\bullet) = \operatorname{im}(\partial_{n+1})$$

- Since $\partial_n \circ \partial_{n+1} = 0$, we have that ∂_n factors through $\operatorname{coker}(\partial_{n+1})$. Then the composite $B_n(C_\bullet) \to C_n \xrightarrow{\partial_n} C_{n-1}$ is 0 since $B_n(C_\bullet) \to C_n \to \operatorname{coker}(\partial_{n+1}) = 0$. By the universal property of kernels, there exists a unique morphism $B_n(C_\bullet) \to Z_n(C_\bullet)$:

$$\begin{tikzcd}
B_n(C_\bullet) \arrow[d, dashed, "\exists!"] \arrow[dr] & & & \\
Z_n(C_\bullet) \arrow[r] & C_n \arrow[r] \arrow[dr, "\partial_n"'] & \operatorname{coker}(\partial_{n+1}) \arrow[d, dashed, "\exists!"] & \\
& & C_{n-1} &
\end{tikzcd}$$

Define the nth homology of the chain complex to be

$$H_n(C_\bullet) = \operatorname{coker}(B_n(C_\bullet) \to Z_n(C_\bullet)) = \operatorname{coker}(\operatorname{im}(\partial_{n+1}) \to \ker(\partial_n))$$

Definition 2.2.2 (The Homology Functor)
Let $\mathcal{A}$ be an abelian category. Let $n \in \mathbb{N}$. Define the nth homology functor $H_n : \mathbf{Ch}(\mathcal{A}) \to \mathcal{A}$ by as follows.

- For a chain complex $C_\bullet$, we have $H_n(C_\bullet)$ is the nth homology of the chain complex.
- For a chain map $C_\bullet \to C'_\bullet$, we obtain maps $Z_n(C_\bullet) \to Z_n(C'_\bullet)$ and $B_n(C_\bullet) \to B_n(C'_\bullet)$ since kernels and images are functorial. Since cokernels are also functorial, we obtain a morphism $H_n(C_\bullet) \to H_n(C'_\bullet)$.

Lemma 2.2.3
Let $\mathcal{A}$ be an abelian category. Then $H_* : \mathbf{Ch}(\mathcal{A}) \to \mathcal{A}$ is an additive functor.

Definition 2.2.4 (Quasi-Isomorphisms) Let $C_\bullet, D_\bullet$ be chain complexes. Let $f : C_\bullet \to D_\bullet$ be a chain map. We say that f is a quasi-isomorphism if the induced map in homology $f_* : H_n(C_\bullet) \to H_n(D_\bullet)$ is an isomorphism for all $n \in \mathbb{N}$.

Example 2.2.5
The chain map given by the vertical arrows in $\mathbf{Ab}$ is a quasi-isomorphism:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d, "0"'] & \mathbb{Z} \arrow[r, "\times 2"] \arrow[d, "0"'] & \mathbb{Z} \arrow[r] \arrow[d, "\text{mod } 2"] & 0 \arrow[d, "0"] \\
0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

In particular, there is no non-zero chain map in the opposite direction.

Because cochain complexes is the same data as a chain complex, we can also define homology groups for cochain complexes. We call them the cohomology groups of the cochain complex.

Definition 2.2.6 (Cohomology)
Let $\mathcal{A}$ be an abelian category. Let $C^\bullet$ be a chain complex. Define the nth cohomology of $C^\bullet$ to be

$$H^n(C^\bullet) = \text{coker}(\text{im}(\delta^n) \to \ker(\delta^{n+1}))$$

The construction of cohomology groups are functorial because the homology groups of a chain complex also are.

Definition 2.2.7 (The Cohomology Functor) Let $\mathcal{A}$ be an abelian category. Let $n \in \mathbb{N}$. Define the nth cohomology functor to be the functor

$$H^n : \mathbf{CCh}(\mathcal{A}) \to \mathcal{A}$$

to consist of the following data.

- For each cochain complex $(C^\bullet, \delta^\bullet)$ over $\mathcal{A}$, define $H^n(C^\bullet, \delta^\bullet)$ to be the nth cohomology group of the cochain complex.
- For each chain map $f^\bullet : (C^\bullet, \delta^\bullet) \to (D^\bullet, \delta^\bullet)$, the induced map

$$f^* : H^n(C^\bullet, \delta^\bullet) \to H^n(C^\bullet, \delta^\bullet)$$

  is defined by $[z] \mapsto [f^n(z)]$.

2.3 Chain Homotopies

Definition 2.3.1 (Chain Homotopy)
Let $\mathcal{A}$ be an abelian category. Let $f_\bullet, g_\bullet : C_\bullet \to C'_\bullet$ be two chain maps in $\mathbf{Ch}(\mathcal{A})$. Then a chain homotopy from f to g is a collection of morphisms

$$\eta_n : C_n \to C'_{n+1}$$

in $\mathcal{A}$ such that

$$g_n - f_n = \partial'_{n+1}\eta_n + \eta_{n-1}\partial_n$$

for all $n \in \mathbb{Z}$. In this case, f and g are said to be chain homotopic and we write $f \simeq g$.

We depict the data of a homotopy as the following diagram in $\mathbf{Ch}(\mathcal{A})$:

$$\begin{tikzcd}
\cdots \arrow[r] & C_{n+1} \arrow[r, "\partial_{n+1}"] \arrow[d, "g_{n+1}-f_{n+1}"] \arrow[dl] & C_n \arrow[r, "\partial_n"] \arrow[d, "g_n-f_n"] \arrow[dl, "\eta_n"'] & C_{n-1} \arrow[r] \arrow[d, "g_{n-1}-f_{n-1}"] \arrow[dl, "\eta_{n-1}"'] & \cdots \arrow[dl] \\
\cdots \arrow[r] & C'_{n+1} \arrow[r, "\partial'_{n+1}"] & C'_n \arrow[r, "\partial'_n"] & C'_{n-1} \arrow[r] & \cdots
\end{tikzcd}$$

Note that the notion of chain homotopy is an equivalence relation on $\text{Hom}_{\mathbf{Ch}(\mathcal{A})}(C, D)$ for any two chain complexes $C, D \in \mathbf{Ch}(\mathcal{A})$.

- If $f : C \to D$ is a chain homotopy equivalence, then $f \simeq f$ via the chain homotopy 0.
- If $f, g : C \to D$ are chain maps such that $f \simeq g$ via the chain homotopy η, then $-\eta$ is a chain homotopy such that $g \simeq f$.
- If $f, g, h : C \to D$ are chain maps such that $f \simeq g$ via η and $g \simeq h$ via ν, then $f \simeq h$ via $\eta + \nu$.

Lemma 2.3.2
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes in $\mathcal{A}$. Let $f, g : C \to D$ be chain maps. If f and g are chain homotopic, then

$$H_*(f) = H_*(g) \in \text{Hom}_{\mathcal{A}}(H_n(C), H_n(D))$$

for all $n \in \mathbb{Z}$.

Proposition 2.3.3
Let $\mathcal{A}$ be an abelian group. Let $f_1, g_1 : C_\bullet \to D_\bullet$ and $f_2, g_2 : D_\bullet \to E_\bullet$ be chain maps in $\mathbf{Ch}(\mathcal{A})$. If f_1 and g_1 are chain homotopic and f_2 and g_2 are chain homotopic, then $f_2 \circ f_1$ is chain homotopic to $g_2 \circ g_1$.

Proof The chain homotopies between f_1 and g_1 imposes the identity

$$\partial\eta + \eta\partial = g_1 - f_1$$

for $\eta : C_\bullet \to D_\bullet$ the given chain homotopy. Similarly, for $\nu : D_\bullet \to E_\bullet$ we have the identity

$$\partial\nu + \nu\partial = g_2 - f_2$$

Then we have that

$$\begin{aligned} g_2 \circ g_1 - f_2 \circ f - 1 &= g_2 \circ g_1 - g_2 \circ f_1 + g_2 \circ f_1 - f_2 \circ f_1 \\ &= g_2(g_1 - f_1) + (g_2 - f_2) \circ f_1 \\ &= g_2(\partial\eta + \eta\partial) + (\partial\nu + \nu\partial) \circ f_1 \\ &= \partial g_2\eta + g_2\eta\partial + \partial\nu f_1 + \nu f_1\partial \\ &= \partial(g_2\eta + \nu f_1) + (g_2\eta + \nu f_1)\partial \end{aligned}$$

Thus $g_2\eta + \nu f_1 : C_n \to E_{n+1}$ would be a valid chain homotopy from $g_2 \circ g_1$ to $f_2 \circ f_1$. ■

Proposition 2.3.4
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes. Suppose that $f, f', g, g' : C \to D$ are chain maps such that $f \simeq f'$ and $g \simeq g'$, then $f + g \simeq f' + g'$.

Definition 2.3.5 (Chain Homotopy Equivalence)
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes in $\mathcal{A}$. Let $f : C \to D$ be a chain map. We say that f is a chain homotopy equivalence if there exists a chain map $g : D \to C$ such that

$$g \circ f \simeq \mathrm{id}_C \quad \text{and} \quad f \circ g \simeq \mathrm{id}_D$$

In this case, we call g the chain homotopy inverse of f.

Lemma 2.3.6
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes in $\mathcal{A}$. Let $f : C \to D$ be a chain homotopy equivalence. Then the following are true.

- The map
$$H_*(f) : H_n(C) \xrightarrow{\cong} H_n(D)$$
is an isomorphism for all $n \in \mathbb{Z}$.
- If g is a chain homotopy inverse of f, then $H_*(g)$ is the inverse of $H_*(f)$.

Proof We know that $b_\bullet \circ a_\bullet \simeq \mathrm{id}_{C_\bullet}$ which means that they induce the same map:

$$b_* \circ a_* = \mathrm{id}_{H_n(C_\bullet)}$$

Similarly the chain homotopies $a_\bullet \circ b_\bullet \simeq \mathrm{id}_{D_\bullet}$ induce the same map

$$a_* \circ b_* : \mathrm{id}_{H_n(D_\bullet)}$$

as the identity. Then these two identities mean that a_* is both injective and surjective. ■

Proposition 2.3.7 Let $\mathcal{A}$ be an abelian category. Then chain homotopy equivalence defines an equivalence relation on all chain complexes in $\mathbf{Ch}(\mathcal{A})$.

Proof Clearly any chain complex is chain homotopy equivalent to itself by the identity map. If $C_\bullet$ and $D_\bullet$ are chain homotopy equivalent by the chain maps $a_\bullet : C_\bullet \to D_\bullet$ and $b_\bullet : D_\bullet \to C_\bullet$, then we have the identities $b_\bullet \circ a_\bullet = \text{id}_{C_\bullet}$ and $a_\bullet \circ b_\bullet = \text{id}_{D_\bullet}$. We can then read them in reverse so that $D_\bullet$ and $C_\bullet$ are chain homotopy equivalence by the maps $b_\bullet$ and $a_\bullet$.

Suppose further that $D_\bullet$ and $E_\bullet$ are chain homotopy equivalent via the maps $u_\bullet : D_\bullet \to E_\bullet$ and $v_\bullet : E_\bullet \to D_\bullet$. Then the maps $u_\bullet \circ a_\bullet$ and $b_\bullet \circ v_\bullet$ give a chain homotopy equivalence between $C_\bullet$ and $E_\bullet$. Indeed, upon composition, we have that they are chain homtopic to the identity maps. ■

Definition 2.3.8 (Contractible Chain Complexes)
Let $\mathcal{A}$ be an abelian category. Let $C \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. We say that C is contractible if the identity map id_C is homotopic to the 0 map.

Example 2.3.9
Let $\mathcal{A}$ be an abelian category. Let $X \in \mathcal{A}$ be an object. Let $C \in \mathbf{Ch}(\mathcal{A})$ be the chain complex defined as follows:

$$\cdots \longrightarrow 0 \longrightarrow X \xrightarrow{\text{id}_X} X \longrightarrow 0 \longrightarrow \cdots$$

where the two non-zero terms are in positions n and $n+1$ for some $n \in \mathbb{Z}$. Then C is contractible.

Proof
Without loss of generality we may take $n = 0$. Define $\eta : C_n \to C_{n+1}$ as follows: If $n \neq 0$, then η_n is the 0 map. Also $\eta_0 = \text{id}_X$. I claim that η gives a homotopy $\text{id}_C \simeq 0$. Indeed, if $n \neq 0, 1$, then we have

$$\partial_{n+1} \circ \eta_n + \eta_{n-1} \circ \partial_n = 0 \circ 0 + 0 \circ 0 = 0 = \text{id}_{C_n} - 0$$

If $n = 0$, then we have

$$\partial_1 \circ \eta_0 + \eta_{-1} \circ \partial_0 = \text{id}_X \circ \text{id}_X + 0 \circ 0 = \text{id}_X = \text{id}_{C_0} - 0$$

and finally if $n = 1$, then we have

$$\partial_2 \circ \eta_1 + \eta_0 \circ \partial_1 = 0 \circ 0 + \text{id}_X \circ \text{id}_X = \text{id}_X = \text{id}_{C_1} - 0$$

hence we are done. ■

2.4 The Homotopy Category of Cochain Complexes

Definition 2.4.1 (Homotopy Category of Cochain Complexes) Let $\mathcal{A}$ be an abelian category. Define the homotopy category of chain complexes

$$K(\mathcal{A})$$

to be the category defined as follows.

- The objects are the objects of $\mathbf{CCh}(\mathcal{A})$.
- The morphisms are homotopy classes of chain maps. Namely, for C, D chain complexes, define
$$\text{Hom}_{K(\mathcal{A})}(C, D) = \frac{\text{Hom}_{\mathbf{Ch}(\mathcal{A})}(C, D)}{\sim}$$
- Composition is given by composition of chain maps.

Define similarly the homotopy categories $K^+(\mathcal{A})$, $K^-(\mathcal{A})$ and $K^b(\mathcal{A})$ for $\mathbf{CCh}^+(\mathcal{A})$, $\mathbf{CCh}^-(\mathcal{A})$ and $\mathbf{CCh}^b(\mathcal{A})$ respectively.

Note: $K(\mathcal{A})$ is in general no longer an abelian category, but it is still a triangulated category cf section 7.

Definition 2.4.2 (The Canonical Functor to the Homotopy Category)
Let $\mathcal{A}$ be an abelian category. Define the canonical functor

$$\mathbf{Ch}(\mathcal{A}) \to K(\mathcal{A})$$

as follows.

- It is the identity map on objects.
- For a chain map $f : C \to D$ the image of f is the homotopy class $[f] : C \to D$ of the chain map.

It is clear that the canonical functor is in general not fully faithful. However, the inclusion functor $\mathcal{A} \to \mathbf{Ch}(\mathcal{A}) \to K(\mathcal{A})$ is fully faithful.

Lemma 2.4.3
Let $\mathcal{A}$ be an abelian category. Then the functor

$$\mathcal{A} \xrightarrow{c_0} \mathbf{Ch}(\mathcal{A}) \to K(\mathcal{A})$$

is fully faithful.

Proposition 2.4.4 Let $\mathcal{A}$ be an abelian category. The homotopy category of cochain complexes satisfy the following universal property.

If $F : \mathbf{Ch}(\mathcal{A}) \to \mathcal{D}$ is a functor that sends chain homotopy equivalences to isomorphisms, then F factors uniquely through $K(\mathcal{A})$:

$$\begin{tikzcd} \mathbf{CCh}(\mathcal{A}) \arrow[r] \arrow[dr, "F"'] & K(\mathcal{A}) \arrow[d, dashed, "\exists!"] \\ & \mathcal{D} \end{tikzcd}$$

Corollary 2.4.5
Let $\mathcal{A}$ be an abelian category. Then the cohomology functors $H_\bullet : \mathbf{Ch}(\mathcal{A}) \to \mathcal{A}$ induces a well defined additive functor from $K(\mathcal{A})$ to $\mathcal{A}$.

Proposition 2.4.6
Let R be a commutative ring. Then the following are true.

- There is a natural isomorphism

$$H_n(-) \cong \mathrm{Hom}_{K({}_R\mathbf{Mod})}(c_n(R), -)$$

  for all $n \in \mathbb{Z}$.
- There is a natural isomorphism

$$H^n(\mathrm{Hom}_{\mathcal{A}}(-, M)) \cong \mathrm{Hom}_{K({}_R\mathbf{Mod})}(-, c_n(M))$$

  for any R-module M.

2.5 Operations of Chain Complexes

Definition 2.5.1 (Shifted Chain Complexes)
Let $\mathcal{A}$ be an abelian category. Let $i \in \mathbb{Z}$. Let $C \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Define the ith shifted chain complex of $C_\bullet$ to be the complex $(C[i], \partial[i])$ defined as follows.
- $C[i]_n = C_{n+i}$.
- $\partial[i]_n = (-1)^i \partial_{n+i}$

Visually, the chain complex is shifted i times to the left. Note that the shifted chain complex has a different chain map if i is odd. This definition is only possible in an additive category.

Definition 2.5.2 (The Shifting Functor)
Let $\mathcal{A}$ be an abelian category. Let $i \in \mathbb{Z}$. Define the ith shifting functor

$$-[i] : \mathbf{Ch}(\mathcal{A}) \to \mathbf{Ch}(\mathcal{A})$$

as follows.
- For a chain complex C, define $C[i]$ to be the ith shifted chain complex.
- For a chain map $f : C \to D$, define $f[i] : C[i] \to D[i]$ by $f[i]_n = f_{n+i}$.

Lemma 2.5.3
Let $\mathcal{A}$ be an abelian category. Then the following are true regarding the shifting functor.
- Let $i, j \in \mathbb{Z}$. Then we have
$$(-[j]) \circ (-[i]) = -[i+j]$$
- Let $i \in \mathbb{Z}$. Then $-[i] : \mathbf{Ch}(\mathcal{A}) \to \mathbf{Ch}(\mathcal{A})$ is a natural isomorphism whose inverse is given by $-[-i]$.
- Let $C \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. For any $i \in \mathbb{Z}$ and $n \in \mathbb{Z}$, we have
$$H_n(C[i]) = H_{n+i}(C)$$

Definition 2.5.4 (Shifted Cochain Complexes)

Definition 2.5.5 (The Shifting Functor)

Definition 2.5.6 (The Cone of a Chain Map)
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes in $\mathcal{A}$. Let $f : C \to D$ be a chain map. Define the cone
$$\text{cone}(f)$$
of f to be the complex defined as follows.
- For each $n \in \mathbb{Z}$, define $\text{cone}(f)_n = C_{n-1} \oplus D_n$.
- For each $n \in \mathbb{Z}$, define the differential by
$$\partial(c, d) = (-\partial_C(c), -\partial_D(d) - f(c))$$

Proposition 2.5.7
Let $\mathcal{A}$ be an abelian category. Let $C, D \in \mathbf{Ch}(\mathcal{A})$ be chain complexes in $\mathcal{A}$. Let $f : C \to D$ be a chain map. Then the following are true regarding the cone of f.
- The following sequence in $\mathbf{Ch}(\mathbf{Ch}(\mathcal{A}))$:
$$0 \longrightarrow D \xrightarrow{\text{incl.}} \text{cone}(f) \xrightarrow{(c,d) \to -c} C[-1] \longrightarrow 0$$
is short exact.
- The connecting homomorphism of the induced long exact sequence from the above short exact sequence is given by $H_\bullet(f)$.
- f is a quasi-isomorphism if and only if $\text{cone}(f)$ is exact.

Definition 2.5.8 (The Cylinder of a Chain Map)

Definition 2.5.9 (The Cone of a Chain Complex)
Let $\mathcal{A}$ be an abelian category. Let $X \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Define the cone of X to be

$$C(X) = \text{cone}(\text{id}_X)$$

Lemma 2.5.10
Let $\mathcal{A}$ be an abelian category. Let $X \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Then $\text{cone}(X)$ is contractible.

2.6 Complexes of Complexes

Let $\mathcal{A}$ be an abelian category. We have seen that $\mathbf{Ch}(\mathcal{A})$ is also an abelian category, and so it makes sense to talk about chain complexes of chain complexes. Let $A_\bullet, B_\bullet, C_\bullet$ be chain complexes in an abelian category $\mathcal{A}$. Let $i : A_\bullet \to B_\bullet$ and $j : B_\bullet \to C_\bullet$ be chain maps in $\mathbf{Ch}(\mathcal{A})$. A short exact sequence of chain complexes is a diagram of the form

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+1} \arrow[r, "d_A"] \arrow[d, "i"] & A_n \arrow[r, "d_A"] \arrow[d, "i"] & A_{n-1} \arrow[r] \arrow[d, "i"] & \cdots \\
\cdots \arrow[r] & B_{n+1} \arrow[r, "d_B"] \arrow[d, "j"] & B_n \arrow[r, "d_B"] \arrow[d, "j"] & B_{n-1} \arrow[r] \arrow[d, "j"] & \cdots \\
\cdots \arrow[r] & C_{n+1} \arrow[r, "d_C"] \arrow[d] & C_n \arrow[r, "d_C"] \arrow[d] & C_{n-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

such that for each n (vertically in the diagram), the sequence

$$0 \longrightarrow A_n \xrightarrow{i} B_n \xrightarrow{j} C_n \longrightarrow 0$$

is a short exact sequence. We write this as

$$0 \longrightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \longrightarrow 0$$

Theorem 2.6.1 (S.E.S. Induces L.E.S.)
Let $\mathcal{A}$ be an abelian category. Suppose that the following is a short exact sequence in $\mathbf{Ch}(\mathcal{A})$:

$$0 \longrightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \longrightarrow 0$$

Then there exists a connecting homomorphism $\partial : H_n(C_\bullet) \to H_{n-1}(A_\bullet)$ such that the following sequence of homology

$$\cdots \longrightarrow H_{n+1}(C_\bullet) \xrightarrow{\partial} H_n(A_\bullet) \xrightarrow{i_*} H_n(B_\bullet) \xrightarrow{j_*} H_n(C_\bullet) \xrightarrow{\partial} H_{n-1}(A_\bullet) \longrightarrow \cdots$$

is an exact sequence. Moreover, the connecting homomorphism is natural in $\mathbf{Ch}(\mathbf{Ch}(\mathcal{A}))$.

Corollary 2.6.2
Let $\mathcal{A}$ be an abelian category. Let the following be a short exact sequence of chain complexes in $\mathcal{A}$:

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

Then the following are true.
- f is a quasi-isomorphism if and only if C is exact.
- g is a quasi-isomorphism if and only if A is exact.

Corollary 2.6.3
Let $\mathcal{A}$ be an abelian category. Let the following be a diagram in $\mathbf{Ch}(\mathbf{Ch}(\mathcal{A}))$:

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r] \arrow[d, "f"'] & Y \arrow[r] \arrow[d, "g"'] & Z \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & 0
\end{tikzcd}$$

If two of f, g, h are quasi-isomorphisms, then so is the third one.

Proof
Since the connecting homomorphism in the s.e.s inducing l.e.s is natural, we obtain the following commutative diagram in $\mathcal{A}$:

$$\begin{tikzcd}
\cdots \arrow[r] & H_{n+1}(Z) \arrow[r] \arrow[d, "H_*(h)"'] & H_n(X) \arrow[r] \arrow[d, "H_*(f)"'] & H_n(Y) \arrow[r] \arrow[d, "H_*(g)"'] & H_n(Z) \arrow[r] \arrow[d, "H_*(h)"] & H_{n-1}(X) \arrow[r] \arrow[d, dashed, "H_*(f)"] & \cdots \\
\cdots \arrow[r] & H_{n+1}(Z') \arrow[r] & H_n(X') \arrow[r] & H_n(Y') \arrow[r] & H_n(Z') \arrow[r] & H_{n-1}(X') \arrow[r] & \cdots
\end{tikzcd}$$

If any two out of f, g, h are quasi-isomorphisms, then the five lemma implies that the last map is an isomorphism on homology. ■

Definition 2.6.4 (Double Complexes)
Let $\mathcal{A}$ be an abelian category. A double complex in $\mathcal{A}$ consists of the following data:
- A family of objects $C_{p,q} \in \mathcal{A}$ for $p, q \in \mathbb{Z}$.
- Horizontal differentials $d^h : C_{p,q} \to C_{p-1,q}$ in $\mathcal{A}$ for $p, q \in \mathbb{Z}$.
- Vertical differentials $d^v : C_{p,q} \to C_{p,q-1}$ in $\mathcal{A}$ for $p, q \in \mathbb{Z}$.

such that the following are true.
- $d^h \circ d^h = 0$.
- $d^v \circ d^v = 0$.
- $d^v \circ d^h = d^h \circ d^v$.

We picture a double complex as follows:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots & C_{p-1,q+1} \arrow[l] \arrow[d, "d^v"'] & C_{p,q+1} \arrow[l, "d^h"'] \arrow[d, "d^v"'] & C_{p+1,q+1} \arrow[l, "d^h"'] \arrow[d, "d^v"] & \cdots \arrow[l] \\
\cdots & C_{p-1,q} \arrow[l] \arrow[d, "d^v"'] & C_{p,q} \arrow[l, "d^h"'] \arrow[d, "d^v"'] & C_{p+1,q} \arrow[l, "d^h"'] \arrow[d, "d^v"] & \cdots \arrow[l] \\
\cdots & C_{p-1,q-1} \arrow[l] \arrow[d] & C_{p,q-1} \arrow[l, "d^h"] \arrow[d] & C_{p+1,q-1} \arrow[l, "d^h"] \arrow[d] & \cdots \arrow[l] \\
 & \vdots & \vdots & \vdots &
\end{tikzcd}$$

where each square is commutative. Double complexes are also graded objects of an abelian category equipeed with differentials that decrease the first and second degrees respectively.

Definition 2.6.5 (The Total Complex of a Double Complex) Let $\mathcal{A}$ be an abelian category where $\oplus$ denotes finite product. Let $(C_{\bullet,\bullet}, d^h, d^v)$ be a double complex. Define the total complex $\text{Tot}^{\oplus}(C)_\bullet$ to be the chain complex in $\mathcal{A}$ constructed as follows.

- For each $n \in \mathbb{Z}$, define
$$\text{Tot}^{\oplus}(C)_n = \bigoplus_{p+q=n} C_{p,q}$$
- For each $n \in \mathbb{Z}$, define $d : \text{Tot}^{\oplus}(C)_n \to \text{Tot}^{\oplus}(C)_{n-1}$ by
$$d = d^v + (-1)^n d^h$$

Definition 2.6.6 (Total Homology of a Double Chain Complex) Let $\mathcal{A}$ be an abelian category. Let $(C_{\bullet,\bullet}, d^h, d^v)$ be a double complex in $\mathcal{A}$. Define the total homology of C to be the homology
$$H_\bullet^{\text{Tot}}(C_{\bullet,\bullet}, d^h, d^v) = H_\bullet(\text{Tot}^{\oplus}(C)_\bullet, d)$$
of the total complex $(\text{Tot}^{\oplus}(C)_\bullet, d)$

Definition 2.6.7 (The Product Total Complex of a Double Complex)
Let $\mathcal{A}$ be an abelian category. Let $(C_{\bullet,\bullet}, d^h, d^v)$ be a double complex. Define the product total complex $\text{Tot}^{\Pi}(C)_\bullet$ to be the chain complex in $\mathcal{A}$ constructed as follows.

- For each $n \in \mathbb{Z}$, define
$$\text{Tot}^{\Pi}(C)_n = \prod_{p+q=n} C_{p,q}$$
- For each $n \in \mathbb{Z}$, define $d : \text{Tot}^{\Pi}(C)_n \to \text{Tot}^{\Pi}(C)_{n-1}$ by
$$d = d^v + (-1)^n d^h$$

Definition 2.6.8 (Tensor Product Double Complex) Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, d^C)$ and $(D_\bullet, d^D)$ be two chain complexes. Define the tensor product of $C_\bullet$ and $D_\bullet$ to be the double complex $C_\bullet \otimes D_\bullet$ with the horizontal differential and vertical differential is defined by
$$d^h = d^C \otimes \text{id}_D \quad \text{and} \quad d^v = (-1)^p \text{id}_C \otimes d^D$$
respectively.

Definition 2.6.9 (Tensor Product Chain Complex) Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, d^C)$ and $(D_\bullet, d^D)$ be two chain complexes. Define the tensor product chain complex of $C_\bullet$ and $D_\bullet$ to be the chain complex
$$(C \otimes D)_\bullet = \text{Tot}^{\oplus}(C_\bullet \otimes D_\bullet)$$

Note that this is different from the tensor product double complex of two chain complexes, although they are closely related. Although the tensor product double complex makes more intuitive sense, the canonical product for $\mathbf{Ch}(\mathcal{A})$ is in fact the tensor product chain complex. Moreover, this product is the symmetric monoidal product of $\mathbf{Ch}(\mathcal{A})$.

3 Derived Functors

3.1 Left and Right Exact Functors

Definition 3.1.1 (Left and Right Exact Functors) Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. Let $0 \to A \to B \to C \to 0$ be an exact sequence in $\mathcal{A}$.

- We say that F is right exact if the sequence
$$F(A) \to F(B) \to F(C) \to 0$$
is exact.
- We say that F is left exact if the sequence
$$0 \to F(A) \to F(B) \to F(C)$$
is exact.

Lemma 3.1.2 Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. Then F is exact if and only if F is left exact and right exact.

Proposition 3.1.3 Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor. Then the following are true.

- F is additive and left exact if and only if F preserves all finite limits.
- F is additive and right exact if and only if F preserves all finite colimits.

Since right adjoints preserves limits, any right adjoint functor between abelian categories must be left exact. Moreover, since limits commutes with limits, the limit functor $\lim_{\mathcal{J}}(-) : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$ is also left exact. Dually, left adjoints and the colimit functor must also be right exact.

Proposition 3.1.4 Let $\mathcal{A}$ be an abelian category. Let $M \in \mathcal{A}$. Then the following are true.

- The covariant functor $\text{Hom}(M, -) : \mathcal{A} \to \mathbf{Ab}$ is left exact.
- The contravariant functor $\text{Hom}(-, M) : \mathcal{A} \to \mathbf{Ab}$ is right exact.

Let $\mathcal{C}$ be a category and let $\mathcal{J}$ be a diagram. Recall that if $\mathcal{C}$ is complete, then the limit can be thought of as a functor $\lim_{\mathcal{J}} : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$. Dually, if $\mathcal{C}$ is cocomplete, the colimit can be thought of as a functor $\text{colim}_{\mathcal{J}} : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$. If $\mathcal{C}$ is an abelian category, it makes sense to ask whether these two functors preserve exact sequences.

Proposition 3.1.5 Let $\mathcal{A}$ be an abelian category. Let $\mathcal{J}$ be a small category. Then the following are true.

- If $\mathcal{A}$ is complete, then the functor $\lim_{\mathcal{J}} : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$ is left exact
- If $\mathcal{A}$ is cocomplete, then the functor $\text{colim}_{\mathcal{J}} : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$ is right exact

Theorem 3.1.6 (The FHHF Theorem)
Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a covariant functor. Then the following are true.

- If F is right exact, then there is a natural transformation $F \circ H_n \Rightarrow H_n \circ F$ for all $n \in \mathbb{N}$.
- If F is left exact, then there is a natural transformation $H_n \circ F \Rightarrow F \circ H_n$ for all $n \in \mathbb{N}$.
- If F is exact, then the above natural transformations isomorphisms and are inverses of each other.

Proposition 3.1.7
Let $\mathcal{A}$ be an abelian category. Let $\mathcal{J}$ be a filtered category. Let $F : \mathcal{J} \to \mathbf{Ch}(\mathcal{A})$ be a functor. Then there is a natural isomorphism
$$\operatorname*{colim}_{\mathcal{J}} H_n \circ F \cong H_n\left(\operatorname*{colim}_{\mathcal{J}} F\right)$$
for all $n \in \mathbb{N}$.

Proof
Since filtered colimits commute with finite limits, we know that the filtered colimit functor is left exact. Hence it is an exact functor. By the FHHF theorem, filtered colimits commute with homology. ■

Example 3.1.8
Since products are coproducts and coproducts are filtered colimits, the above prp shows that homology commutes with products.

3.2 Injective and Projective Objects

Injectivity and Projectivity objects are created just for the sake of allowing the Hom functor to be exact. Therefore its definition is also direct.

Definition 3.2.1 (Projective and Injective Objects) Let $\mathcal{A}$ be an abelian category. Let $Y \in \mathcal{A}$ be an object.
- We say that Y is injective if the functor $\mathrm{Hom}(-, Y) : \mathcal{A} \to \mathbf{Ab}$ is exact.
- We say that Y is projective if the functor $\mathrm{Hom}(Y, -) : \mathcal{A} \to \mathbf{Ab}$ is exact.

Lemma 3.2.2
Let $\mathcal{A}$ be an abelian category. Then the following are true.
- Let $I \in \mathcal{A}$ be an objects. Then I is injective if and only if for all objects $X, Y \in \mathcal{A}$ and injective morphisms $f : X \to Y$ and $u : X \to I$, there exists a morphism $g : Y \to I$ such that the following diagram commutes:
$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "f"'] & I \\ Y \arrow[ur, dashed, "\exists"'] & \end{tikzcd}$$
- Let $P \in \mathcal{A}$ be an objects. Then P is projective if and only if for all objects $X, Y \in \mathcal{A}$ and surjective morphisms $f : X \to Y$ and $u : P \to Y$, there exists a morphism $g : P \to X$ such that the following diagram commutes:
$$\begin{tikzcd} & X \arrow[d, "f"] \\ I \arrow[ur, dashed, "\exists"] \arrow[r, "u"'] & Y \end{tikzcd}$$

Lemma 3.2.3 Every free module is projective.

Proof Let R be a ring and let F be a free R-module. Suppose that F has basis B. Let M, N be R-modules. Suppose that $f : N \to M$ is surjective and there exists an R-module homomorphism $g : R^n \to M$. Since f is surjective, for each $b \in B$, we can choose a pre-image for $g(b)$ in N for all $b \in B$. Call it n_b. Now define $h : F \to N$ by $b \mapsto n_b$ and then extend it R-linearly. Now if

$\sum_{b\in B} k_b b \in F$, we have that

$$\begin{aligned}
(f\circ h)\left(\sum_{b\in B} k_b b\right) &= f\left(\sum_{b\in B} k_b h(b)\right) \\
&= f\left(\sum_{b\in B} k_b n_b\right) \\
&= f\left(\sum_{b\in B} k_b n_b\right) \\
&= \sum_{b\in B} k_b f(n_b) \\
&= \sum_{b\in B} k_b g(b) \\
&= g\left(\sum_{b\in B} k_b b\right)
\end{aligned}$$

so that $f\circ h = g$. Thus F is projective. ■

Proposition 3.2.4 Let R be a ring. Let P be an R-module. Then the following are equivalent.

- P is projective
- For every surjective homomorphism $f : N \twoheadrightarrow M$ and every R-module homomorphism $g : P \to M$, there exists a module homomorphism $h : P \to N$ such that the following diagram commutes:

$$\begin{tikzcd}
 & N \arrow[d, two heads, "f"] \\
P \arrow[ur, dashed, "\exists h"] \arrow[r, "g"'] & M
\end{tikzcd}$$

- $P \oplus Q$ is a free R-module for some R-module Q.

Proof

- (3) $\implies$ (1): Suppose that there exists a module Q such that $P\oplus Q$ is free. Let $f : N \to M$ be a surjective R-module homomorphism and let $g : P \to M$ be an R-module homomorphism.

■

Proposition 3.2.5 Let R be a ring. Let P_i be an R-module for each $i \in I$. Then the direct sum $\bigoplus_{i\in I} P_i$ is projective if and only if each P_i is projective.

Example 3.2.6 Let k be a field. Then every k-module is injective.

Example 3.2.7 Let R be a DVR. Let t be a uniformizing parameter of R. Then $R[1/t]/R$ is an injective R-module.

Proposition 3.2.8 (Baer's Criterion)
Let R be a ring. Let M be a left R-module. Then M is injective if and only if for every left ideal I of R and R-module homomorphism $\phi : I \to M$, there exists a map $\overline{\phi}$ such that the following diagram commutes:

$$\begin{tikzcd} I \arrow[r, "\phi"] \arrow[d, hook] & M \\ R \arrow[ur, dashed, "\exists\overline{\phi}"'] & \end{tikzcd}$$

Definition 3.2.9 (Enough Injectives and Enough Projectives) Let $\mathcal{A}$ be an abelian category. $\mathcal{A}$ is said to have enough injectives if every object is the subobject of an injective object. $\mathcal{A}$ is said to have enough projectives if every object is the quotient of an projective object.

There are however equivalent definitions from the categorical point of view.

3.3 Resolutions of Objects

There are in general, four types of resolutions. Namely injective resolutions, projective resolutions, free resolutions and acyclic resolutions. We will study all four of them and their relations in this section.

Definition 3.3.1 (Injective Resolution)
Let $\mathcal{A}$ be an abelian category. Let $A \in \mathcal{A}$. An injective resolution of A is a cochain complex $I \in \mathbf{CCh}_{\geq 0}(\mathcal{A})$ such that the following are true.

- I^n is injective for all $n \geq 0$.
- There exists a quasi-isomorphism $\iota : c_0(A) \to I$ depicted as

$$\begin{tikzcd} \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, "\iota"] & 0 \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & 0 \arrow[r] & I^0 \arrow[r] & I^1 \arrow[r] & \cdots \end{tikzcd}$$

Example 3.3.2
An injective resolution of $\mathbb{Z}$ in $\mathbf{Ab}$ is given by

$$\begin{tikzcd} \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d, "\text{incl.}"] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] & \cdots \\ \cdots \arrow[r] & 0 \arrow[r] & \mathbb{Q} \arrow[r] & \mathbb{Q}/\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots \end{tikzcd}$$

Proposition 3.3.3 Let $\mathcal{A}$ be an abelian category. Then $\mathcal{A}$ has enough injectives if and only if every object of $\mathcal{A}$ has an injective resolution.

Proposition 3.3.4 Let $\phi : A \to A'$ be a morphism in an abelian category $\mathcal{A}$. Suppose that there are injective resolutions

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\phi"'] & I^0 \arrow[r] & I^1 \arrow[r] & \cdots \\ 0 \arrow[r] & A' \arrow[r] & J^0 \arrow[r] & J^1 \arrow[r] & \cdots \end{tikzcd}$$

for A and A' respectively. Then there exists a chain map extending ϕ such that the following diagram commutes:

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\phi"'] & I^0 \arrow[r] \arrow[d, "\phi^0"'] & I^1 \arrow[r] \arrow[d, "\phi^1"'] & \cdots \\ 0 \arrow[r] & A' \arrow[r] & J^0 \arrow[r] & J^1 \arrow[r] & \cdots \end{tikzcd}$$

Moreover, any two such chain maps are homotopic.

Lemma 3.3.5 Let $\mathcal{A}$ be an abelian category. Then any two injective resolutions of an object A are homotopically equivalent.

Definition 3.3.6 (Projective Resolution)
Let $\mathcal{A}$ be an abelian category. Let $A \in \mathcal{A}$. A projective resolution of A is a chain complex $P \in \mathbf{Ch}_{\geq 0}(\mathcal{A})$ such that the following are true.

- P_n is injective for all $n \geq 0$.
- There exists a quasi-isomorphism $\iota : P \to c_0(A)$ depicted as

$$\begin{tikzcd}
\cdots \arrow[r] & P_1 \arrow[r] \arrow[d] & P_0 \arrow[r] \arrow[d, "\iota"'] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & A \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Example 3.3.7
Let R be a commutative ring. An projective resolution of R in ${}_R\mathbf{Mod}$ is given by

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & R[x] \arrow[r, "\cdot x"] \arrow[d] & R[x] \arrow[r] \arrow[d, "\mathrm{ev}_0"] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & R \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Example 3.3.8
Let R be a commutative ring. An projective resolution of R in ${}_R\mathbf{Mod}$ is given by

$$\begin{tikzcd}
\cdots \arrow[r] & \frac{R[x]}{(x^2)} \arrow[r, "\cdot x"] \arrow[d] & \frac{R[x]}{(x^2)} \arrow[r, "\cdot x"] \arrow[d] & \frac{R[x]}{(x^2)} \arrow[r] \arrow[d, "\mathrm{ev}_0"] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & R \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Proposition 3.3.9 Let $\mathcal{A}$ be an abelian category. Then $\mathcal{A}$ has enough projectives if and only if every object of $\mathcal{A}$ has a projective resolution.

Proposition 3.3.10 Let $\phi : A \to A'$ be a morphism in an abelian category $\mathcal{A}$. Suppose that there are projective resolutions

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] & P_1 \arrow[r] & A \arrow[r] \arrow[d, "\phi"'] & 0 \\
\cdots \arrow[r] & Q_2 \arrow[r] & Q_1 \arrow[r] & A' \arrow[r] & 0
\end{tikzcd}$$

for A and A' respectively. Then there exists a chain map extending ϕ such that the following diagram commutes:

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] \arrow[d, "\phi_2"'] & P_1 \arrow[r] \arrow[d, "\phi_1"'] & A \arrow[r] \arrow[d, "\phi"'] & 0 \\
\cdots \arrow[r] & Q_2 \arrow[r] & Q_1 \arrow[r] & A' \arrow[r] & 0
\end{tikzcd}$$

Moreover, any two such chain maps are homotopic.

Lemma 3.3.11 Let $\mathcal{A}$ be an abelian category. Then any two projective resolutions of an object A are homotopically equivalent.

3.4 Derived Functors

Definition 3.4.1 (Right Derived Functors)
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough injective. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Define the right derived functors

$$R^i F : \mathcal{A} \to \mathcal{B}$$

of F for $i \geq 0$ as follows.

- On objects, $R^i F(A) = H^i(F(I^\bullet))$ where $d : A \to I^\bullet$ is an injective resolution of A
- On Morphisms, $R^i F(\phi : A \to B) = H^i(F(\phi^\bullet : I^\bullet \to (I')^\bullet))$ where $\phi^\bullet : I^\bullet \to (I')^\bullet$ is an extension of ϕ to resolutions.

Proposition 3.4.2
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough injective. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Then the following are true.

- $R^0 F = F$.
- The nth right derived functor $R^n F$ is an additive functor from $\mathcal{A}$ to $\mathcal{B}$.
- If $A \in \mathcal{A}$ is an injective object, then $R^n F(A) = 0$ for all $n > 0$.

Proposition 3.4.3
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough injective. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. For any short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

there is a canonical long exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow R^1(A) \longrightarrow R^1(B) \longrightarrow R^1(C) \longrightarrow R^2(A) \longrightarrow \cdots$$

Definition 3.4.4 (Left Derived Functors)
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough projectives. Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor. Define the left derived functors

$$L_i F : \mathcal{A} \to \mathcal{B}$$

of F $i \geq 0$ as follows.

- On objects, $L_i F(A) = H_i(F(P^\bullet))$ where $d : P_\bullet \to A$ is an projective resolution of A
- On Morphisms, $L_i F(\phi : A \to B) = L_i(F(\phi_\bullet : P_\bullet \to (P')_\bullet))$ where $\phi_\bullet : P_\bullet \to (P')_\bullet$ is an extension of ϕ to resolutions.

Proposition 3.4.5
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough projectives. Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor. Then the following are true.

- $L_0 F = F$.
- The nth left derived functor $L_n F$ is an additive functor from $\mathcal{A}$ to $\mathcal{B}$.
- If $A \in \mathcal{A}$ is a projective object, then $L_n F(A) = 0$ for all $n > 0$.

Proposition 3.4.6
Let $\mathcal{A}, \mathcal{B}$ be abelian categories such that $\mathcal{A}$ has enough projectives. Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor. For any short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

there is a canonical long exact sequence

$$\cdots \longrightarrow L_2(C) \longrightarrow L_1(A) \longrightarrow L_1(B) \longrightarrow L_1(C) \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

Definition 3.4.7 (The Ext Functor)
Let $\mathcal{A}$ be an abelian category. Let $A \in \mathcal{A}$.

- Suppose that $\mathcal{A}$ has enough injectives. Define the nth Ext functor of A to be the right derived functor
$$\text{Ext}^n_{\mathcal{A}}(A, -) = R^n(\text{Hom}(A, -))$$
- Suppose that $\mathcal{A}$ has enough projectives. Define the nth Ext functor of A to be the right derived functor
$$\text{Ext}^n_{\mathcal{A}}(-, A) = R^n(\text{Hom}(-, A))$$

TBA: Functorial in A.

Proposition 3.4.8
Let $\mathcal{A}$ be an abelian category with enough injectives and projectives. Then there is an isomorphism

$$R^n(\text{Hom}(A, -))(B) \cong R^n(\text{Hom}(-, B))(A)$$

that is natural in A and B for all $n \in \mathbb{N}$.

3.5 δ-Functors

Definition 3.5.1 (δ-Functors) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. A homological δ-functor is a collection $\{T_n : \mathcal{A} \to \mathcal{B} \mid n \in \mathbb{N}\}$ of additive functors such that for any short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

there are morphisms $\delta_n : T_n(C) \to T_n(A)$ for $n \in \mathbb{N}$ such that the following are true.

- There is a long exact sequence
$$\cdots \longrightarrow T_{n+1}(C) \xrightarrow{\delta_{n+1}} T_n(A) \longrightarrow T_n(B) \longrightarrow T_n(C) \xrightarrow{\delta_n} T_{n-1}(A) \longrightarrow \cdots$$
- If there is a morphism of short exact sequences
$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] & C \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & 0 \end{tikzcd}$$
the following diagram commutes:
$$\begin{tikzcd} T_n(C) \arrow[r, "\delta_n"] \arrow[d] & T_{n-1}(A) \arrow[d] \\ T_n(C') \arrow[r, "\delta'_n"] & T_{n-1}(A') \end{tikzcd}$$

3.6 The Ext Functor of ${}_R\text{Mod}$ and Extensions

Definition 3.6.1 (Extensions) Let R be a ring. Let M, N be R-modules. An extension of M by N is a short exact sequence

$$0 \longrightarrow N \longrightarrow E \longrightarrow M \longrightarrow 0$$

of R-modules.

Definition 3.6.2 (Equivalent Extensions) Let R be a ring. Let M, N be R-modules. We say that two extensions

$$0 \longrightarrow N \longrightarrow E \longrightarrow M \longrightarrow 0$$

$$0 \longrightarrow N \longrightarrow F \longrightarrow M \longrightarrow 0$$

are equivalent if there exists an R-module homomorphism $\phi : E \to F$ such that the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r] \arrow[d, "\mathrm{id}_N"] & E \arrow[r] \arrow[d, "\phi"] & M \arrow[r] \arrow[d, "\mathrm{id}_M"] & 0 \\
0 \arrow[r] & N \arrow[r] & F \arrow[r] & M \arrow[r] & 0
\end{tikzcd}$$

Proposition 3.6.3 Let R be a ring. Let M, N be R-modules. Suppose that the following two extensions are equivalent:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r] \arrow[d, "\mathrm{id}_N"] & E \arrow[r] \arrow[d, "\phi"] & M \arrow[r] \arrow[d, "\mathrm{id}_M"] & 0 \\
0 \arrow[r] & N \arrow[r] & F \arrow[r] & M \arrow[r] & 0
\end{tikzcd}$$

Then ϕ is an isomorphism. Moreover, equivalent extensions is an equivalence relation.

Definition 3.6.4 (Split Extensions) Let R be a ring. Let M, N be R-modules. We say that an extension splits if it is equivalent to the following extension

$$0 \longrightarrow N \longrightarrow N \oplus M \longrightarrow M \longrightarrow 0$$

of R-modules. The above extension is called a trivial extension.

Proposition 3.6.5 Let R be a ring. Let M, N be R-modules. There is a bijection

$$\frac{\{\text{Extensions of } M \text{ by } N\}}{\cong} \quad \overset{1:1}{\longleftrightarrow} \quad \text{Ext}^1_R(M, N)$$

where $\cong$ means equivalence of extensions. Moreover, the trivial extension corresponds to the zero element of $\text{Ext}^1_R(M, N)$.

Proposition 3.6.6 Let A, B be R-modules. Then the following are true regarding the Ext group.

- $\text{Ext}^0_R(A, B) \cong \text{Hom}_R(A, B)$
- $\text{Ext}^i_R(A, B) = 0$ for all $i > 0$ if A is projective or B is injective
- $\text{Ext}^i_R(A, B) = 0$ for all $i \geq 2$ if A, B are $\mathbb{Z}$-modules.

TBA: External products

3.7 The Tor Functor of ${}_R\mathbf{Mod}$ and Flat Modules

Let R be a ring. Recall that the tensor product is a bifunctor

$$- \otimes_R - : {}_R\mathbf{Mod} \times \mathbf{Mod}_R \to \mathbf{Ab}$$

Definition 3.7.1 (The Tor Functor)
Let R be a ring. Let $n \in \mathbb{N}$.

- Let $B \in \mathbf{Mod}_R$. Define the nth tor functor of B to be

$$\text{Tor}_R^n(-, B) = L_n(- \otimes_R B)$$

- Let $A \in {}_R\mathbf{Mod}$. Define the nth tor functor of A to be

$$\text{Tor}_R^n(A, -) = L_n(A \otimes_R -)$$

TBA: Functoriality of Tor construction.

Proposition 3.7.2
Let R be a ring. Let $A \in {}_R\mathbf{Mod}$ and $B \in \mathbf{Mod}_R$. Then there is an isomorphism

$$\text{Tor}_i^R(A, B) = L_i(- \otimes_R B)(A) \cong L_i(A \otimes_R -)(B)$$

that is natural in A and B for all $i \in \mathbb{N}$.

While projective modules allows the exactness of $\text{Hom}(M, -)$, flat modules allows the exactness of $- \otimes_R M$.

Definition 3.7.3 (Flat Modules) Let R be a ring. An R-module M is said to be flat if for every injective linear map $\phi : K \to L$ of R-modules, the map

$$\phi \otimes \text{id}_M : K \otimes_R M \to L \otimes_R M$$

is also injective.

Proposition 3.7.4 Let R be a ring and M an R-module. Then M is a flat module if and only if for every short exact sequence $0 \to K \to L \to J \to 0$, the sequence

$$0 \to K \otimes_R M \to L \otimes_R M \to J \otimes_R M \to 0$$

is also exact.

Proposition 3.7.5 Let R be a ring. Then the following are true.

- Product: If A and B are flat over R then $A \otimes_R B$ is flat over R
- Base Change: Let S be an R-algebra ($R \to S$ a ring hom). Then $M \otimes_R S$ is flat over S for any flat R-module M
- Transitivity: Let S be an R-algebra such that S is flat over R. If C is flat over S then C is flat over R.

We have the following inclusion of modules

$$\text{Free Modules} \subset \text{Projective Modules} \subset \text{Flat Modules} \subset \text{Torsion Free Modules}$$

It is important to note that the "duality" between projective and injective modules is given from the bifunctor $\text{Hom}(-, -)$ while the "duality" given between projective and flat modules is given by the tensor-hom adjunction.

TBA: Pairing

4 Homological Applications in Algebra

4.1 Divisible Groups

(can be removed to Infinite Abelian Groups)

Definition 4.1.1 (Divisible Groups) Let A be an abelian group. Let $n \in \mathbb{N} \setminus \{0\}$. We say that A is n-divisible if for all $g \in A$, there exists $h \in A$ such that $nh = g$.

Proposition 4.1.2 Let A be an abelian group. Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are equivalent.
- A is n-divisible.
- $nA = A$.
- A is an injective $\mathbb{Z}$-module.

Definition 4.1.3 (The Prüfer p-Group) Let p be a prime. Define the Prüfer p-Group to be the quotient group

$$\mathbb{Z}(p^\infty) = \frac{\mathbb{Z}[1/p]}{\mathbb{Z}}$$

Example 4.1.4 The following are divisible groups.
- $\mathbb{Q}$.
- $\mathbb{Z}[1/p]/\mathbb{Z}$.

Theorem 4.1.5 (Structure Theorem for Divisible Groups) Let A be an abelian group. If A is divisible, then there is an isomorphism

$$A \cong \left(\bigoplus_{p \text{ is prime}} \frac{\mathbb{Z}[1/p]}{\mathbb{Z}}^{k_p} \right) \oplus \mathbb{Q}^I$$

for some choice of indexing sets I and some $k_p \in \mathbb{N}$.

4.2 Koszul Complexes

Definition 4.2.1 (Koszul Complex)
Let R be a ring. Let M be a left R-module. Let $\varphi : M \to R$ be an R-module homomorphism. Define the Koszul complex $K_\bullet(\varphi)$ of φ to be the chain complex given as follows.
- For each $n \in \mathbb{N}$, define
$$K_n(\varphi) = \bigwedge_{i=1}^n M$$
and $K_n(\varphi) = 0$ for $n < 0$.
- Define the differential $d : K_n(M) \to K_{n-1}(M)$ by
$$d(m_1 \wedge \cdots \wedge m_n) = \sum_{i=1}^n (-1)^{i+1} \varphi(m_i) m_1 \wedge \cdots \wedge \widehat{m_i} \wedge \cdots \wedge m_n$$
for $m_1, \dots, m_n \in M$ and then extend it R-linearly.

When M is a finitely generated free R-module, it suffices to see where a set of R-basis is mapped to.

Definition 4.2.2 (Koszul Complex on Central Elements)
Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Define the Koszul complex on

$x_1, \dots, x_n$ to be

$$K_\bullet(x_1, \dots, x_n) = K\left((r_1, \dots, r_n) \in R^n \mapsto \sum_{i=1}^n r_i x_i \in R\right)$$

Let R be a ring. Let $x_1, \dots, x_n \in R$ be central elements. Let $e_1, \dots, e_n$ be an R-basis for R^n. Then we may describe the differential on the R-basis of $\Lambda^k(R^n)$ as follows:

$$d(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{j=1}^k (-1)^{j+1} x_j e_{i_1} \wedge \cdots \wedge \widehat{e_{i_j}} \wedge \cdots \wedge e_{i_k}$$

Example 4.2.3
Let R be a ring. Let $x \in Z(R)$. Then the Koszul complex $K_\bullet(x)$ is the chain complex:

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow 0$$

Lemma 4.2.4
Let R be a ring. Let $x, y \in Z(R)$ be central elements. Then we have

$$K_\bullet(x, y) = \text{Tot}^\oplus\left(K_\bullet(x) \otimes_R K_\bullet(y)\right)$$

Lemma 4.2.5
Let R be a commutative ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let $A = (a_{ij}) \in GL_n(R)$ be an invertible matrix. Then there is an isomorphism

$$K_\bullet(x_1, \dots, x_n) \cong K_\bullet\left(\sum_{i=1}^n a_{1i} f_i, \dots, \sum_{i=1}^n a_{ni} f_i\right)$$

For an example, let R be a commutative ring. Let $x, y \in R$. Then the Koszul complex $K(x, y)$ is given by

$$0 \longrightarrow R^2 \wedge R^2 \longrightarrow R^2 \longrightarrow R \longrightarrow 0$$

The differentials are given as follows.

- The first differential $R^2 \to R$ is given by $(r, s) \mapsto rx + sy$. It can also be given as a 1×2 matrix as $\begin{pmatrix} x & y \end{pmatrix}$. Also alternatively, we can write an R-basis for R^2 with $(1, 0)$ and $(0, 1)$. Then define the map $R^2 \to R$ by $(1, 0) \mapsto x$ and $(0, 1) \mapsto y$.
- Now recall that $R^2 \wedge R^2$ has R-basis given by $(1, 0) \wedge (0, 1)$ and so there is an isomorphism $R^2 \wedge R^2 \cong R$. Then the second differential $R \to R^2$ is given by $1 \mapsto (x, -y)$.

Definition 4.2.6 (Koszul (Co)Homology) Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be an R-module. Define the Koszul homology of M with respect to $x_1, \dots, x_n$ by

$$H_k^{\text{Kos}}(x_1, \dots, x_n; M) = H_k(K(x_1, \dots, x_n; M))$$

Define the Koszul cohomology of M with respect to the central elements by

$$H^k_{\text{Kos}}(x_1, \dots, x_n; M) = H^k(\text{Hom}_R(K(x_1, \dots, x_n)), M)$$

Lemma 4.2.7 Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be an R-module. Then the following are true.

- $H_0^{\text{Kos}}(x_1, \dots, x_n; M) = \frac{M}{(x_1, \dots, x_n)M}$
- $H^0_{\text{Kos}}(x_1, \dots, x_n; M) = \{m \in M \mid x_i m = 0 \text{ for all } i\}$

Theorem 4.2.8 (Kunneth Theorem) Let R be a ring. Let $x_1, \dots, x_n \in R$ be central elements. Let $C_\bullet$ be a chain complex of R-modules. Then there is an exact sequence given by

$$0 \longrightarrow H_0(x_1, \dots, x_n; H_q(C_\bullet)) \longrightarrow H_q^{\text{Tot}}(K(x_1, \dots, x_n) \otimes_R C_\bullet) \longrightarrow H_1(x_1, \dots, x_n; H_{q-1}(C_\bullet)) \longrightarrow 0$$

Theorem 4.2.9 (Self Duality of the Koszul Complex) Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be a left R-module. Then there is a natural isomorphism of chain complexes

$$\text{Hom}(K(x_1, \dots, x_n); M) \cong K(x_1, \dots, x_n; M)$$

The self duality of the Koszul complex means that we could have equally defined the differential to be going up the chain instead of down. In this case, the complex can be defined as

$$0 \longrightarrow \bigwedge_{i=1}^n R^n \xrightarrow{d_n} \bigwedge_{i=1}^{n-1} R^n \longrightarrow \cdots \longrightarrow R^n \xrightarrow{d_1} R \longrightarrow 0$$

where the differential $d_k : \bigwedge_{i=1}^{k-1} R^n \to \bigwedge_{i=1}^k R^n$ is now defined by

$$d(e_{i_1} \wedge \cdots \wedge e_{i_{k-1}}) = \sum_{j=1}^n x_j e_j \wedge e_{i_1} \wedge \cdots \wedge e_{i_{k-1}}$$

Corollary 4.2.10 Let R be a ring. Let $x_1, \dots, x_n \in Z(R)$ be central elements. Let M be a left R-module. Then there is an isomorphism

$$H_p^{\text{Kos}}(x_1, \dots, x_n; M) \cong H_{\text{Kos}}^{n-p}(x_1, \dots, x_n; M)$$

4.3 Local Cohomology

5 Spectral Sequences

5.1 General Spectral Sequences

Definition 5.1.1 (Homological Spectral Sequences)
Let $\mathcal{A}$ be an abelian category. Let $r_0 \in \mathbb{N}$. A homological spectral sequence $(E^r_{\bullet,\bullet}, d^r_{\bullet,\bullet})_{r \geq r_0}$ starting on the r_0th page consists of the following data.

- For each $r \geq r_0$, a $(\mathbb{Z}, \mathbb{Z})$-bigraded object $E^r_{\bullet,\bullet}$ called the rth page.
- For all $p, q \in \mathbb{Z}$ and $r \geq r_0$, a morphism
$$d^r_{p,q} : E^r_{p,q} \to E^r_{p-r,q+r-1}$$
such that $d^r \circ d^r = 0$ called the differential.
- For each $p, q \in \mathbb{Z}$ and $r \geq r_0$, an isomorphism
$$E^{r+1}_{p,q} = \frac{\ker(d^r : E^r_{p,q} \to E^r_{p-r,q+r-1})}{\operatorname{im}(d^r : E^r_{p+r,q-r+1} \to E^r_{p,q})}$$

We say that the total degree of $E^r_{p,q}$ is $n = p + q$.

Definition 5.1.2 (Cycle and Boundary Notation)
Let $\mathcal{A}$ be an abelian category. Let $(E^r_{\bullet,\bullet}, d^r)_{r \geq r_0}$ be a spectral sequence.

- For each $p, q \in \mathbb{Z}$, define $Z^r_{p,q} = \ker(d : E^r_{p,q} \to E^r_{p-r,q+r-1})$.
- For each $p, q \in \mathbb{Z}$, define $B^r_{p,q} = \operatorname{im}(d : E^r_{p+r,q-r+1} \to E^r_{p,q})$.

Definition 5.1.3 (Cohomological Spectral Sequences) Let $\mathcal{A}$ be an abelian category. A cohomological spectral sequence consists of the following data.

- A collection of objects $E^{\bullet,\bullet}_r = \{E^{p,q}_r \in \mathcal{A} \mid p, q \in \mathbb{Z}\}$ called pages for each $r \in \mathbb{N}$. So that there is a sequence
$$E^{\bullet,\bullet}_1, E^{\bullet,\bullet}_2, E^{\bullet,\bullet}_3, \dots$$
of family of objects
- A degree (p, q) map
$$d^{p,q}_r : E^{p,q}_r \to E^{p-r,q+r-1}_r$$
for each $p, q \in \mathbb{Z}$ and $r \in \mathbb{N}$ such that $d_r \circ d_r = 0$
- Isomorphisms of the form $E^{\bullet,\bullet}_{r+1} = H^\bullet(E^{\bullet,\bullet}_r, d_r)$. This means that
$$E^{p,q}_{r+1} = \frac{\ker(d_r : E^{p,q}_r \to E^{p-r,q+r-1}_r)}{\operatorname{im}(d_r : E^{p+r,q-r+1}_r \to E^{p,q}_r)}$$

Notice that cohomological spectral sequences are really the same thing as homological spectral sequences, just reindex the objects by $E^{p,q}_r = E^r_{-p,-q}$.

Definition 5.1.4 (The Infinity Terms)
Let $\mathcal{A}$ be an abelian category. Let $(E^r_{\bullet,\bullet}, d^r)_{r \geq r_0}$ be a spectral sequence.

- For each $p, q \in \mathbb{Z}$, define
$$Z^\infty_{p,q} = \lim_{r \to \infty} Z^r_{p,q}$$
if the limit exists.
- For each $p, q \in \mathbb{Z}$, define
$$B^\infty_{p,q} = \operatorname{colim}_{r \to \infty} B^r_{p,q}$$
if the colimit exists.
- For each $p, q \in \mathbb{Z}$, define the infinity term to be
$$E^\infty_{p,q} = \operatorname{coker}(B^\infty_{p,q} \to Z^\infty_{p,q})$$
if the colimit and limit exists.

If $\mathcal{A}$ is a concrete category, then the maps $Z_{p,q}^{r+1} \to Z_{p,q}^r$ and $B_{p,q}^r \to B_{p,q}^{r+1}$ are inclusion maps so that they are subsets of $E_{p,q}^{r_0}$. Then the limit and colimit evaluates to be the intersection and union respectively. Hence

$$E_{p,q}^\infty = \frac{\bigcap_{r=r_0}^\infty Z_{p,q}^r}{\bigcup_{r=r_0}^\infty B_{p,q}^r}$$

5.2 Stability of Spectral Sequences

Definition 5.2.1 (Stable Spectral Sequences)
Let $\mathcal{A}$ be an abelian category. Let $(E_{\bullet,\bullet}^r, d^r)_{r \geq r_0}$ be a spectral sequence. We say that the spectral sequence is stable at $s \geq r_0$ if there is an isomorphism

$$E_{p,q}^\infty \cong E_{p,q}^s$$

for all $p, q \in \mathbb{Z}$.

Definition 5.2.2 (Bounded Spectral Sequences)
Let $\mathcal{A}$ be an abelian category. Let $(E_{\bullet,\bullet}^r, d^r)_{r \geq r_0}$ be a spectral sequence.

- We say that the spectral sequence is bounded if for each $n \in \mathbb{Z}$, there are only finitely many non-zero terms of total degree n in E^r for all $r \in \mathbb{N}^+$.
- We say that the spectral sequence is first quadrant if $E_{p,q}^r = 0$ for all $p < 0$ or $q < 0$.

Clearly if a homological spectral sequence is first quadrant, then it is bounded.

Lemma 5.2.3
Let $(E_{\bullet,\bullet}^r, d^r)$ be a bounded homological spectral sequence. Then the following are true.

- For each $(p, q) \in \mathbb{Z}^2$, there exists $s \in \mathbb{N}$ such that $E_{p,q}^r \cong E_{p,q}^s$ for all $r \geq s$.
- The spectral sequence is stable.

Definition 5.2.4 (Collapsing Spectral Sequences)
Let $\mathcal{A}$ be an abelian category. Let $(E_{\bullet,\bullet}^r, d^r)_{r \geq r_0}$ be a spectral sequence. We say that it collapses at $s \geq r_0 \geq 2$ if there is exactly one non-zero row or column in $E_{\bullet,\bullet}^s$.

Lemma 5.2.5
Let $\mathcal{A}$ be an abelian category. Let $(E_{\bullet,\bullet}^r, d^r)_{r \geq r_0}$ be a spectral sequence. If the spectral sequence collapses at $s \geq r_0$, then it is stable at s.

5.3 Convergence

Definition 5.3.1 (Weakly Convergent)
Let $\mathcal{A}$ be an abelian category. Let $(E_{\bullet,\bullet}^r, d^r)_{r \geq r_0}$ be a spectral sequence in $\mathcal{A}$. Let $H_\bullet$ be a $\mathbb{Z}$-graded object in $\mathcal{A}$. Let $(F_m H_n)_{m \geq r_0}$ be a filtration for H_n for each n.

- We say that the spectral sequence weakly converges to $H_\bullet$ if the filtrations are exhaustive and there are isomorphisms

$$E_{p,q}^\infty \cong \frac{F_p H_{p+q}}{F_{p-1} H_{p+q}}$$

  We denote it by $E_{p,q}^r \Rightarrow H_\bullet$.
- We say that the spectral sequence converges to $H_\bullet$ if it is weakly convergent to $H_\bullet$ and the filtration is Hausdorff.

5.4 The Spectral Sequence of Exact Couples

Definition 5.4.1 (Exact Couple)
Let $\mathcal{A}$ be an abelian category. An exact couple in $\mathcal{A}$ consists of the following:
- Two objects $A, E \in \mathcal{A}$.
- Morphisms $i : A \to A$ and $j : A \to E$ and $k : E \to A$ in $\mathcal{A}$.

such that the following diagram commutes and is exact:

$$\begin{tikzcd} A \arrow[rr, "i"] & & A \arrow[dl, "j"] \\ & E \arrow[ul, "k"] & \end{tikzcd}$$

We denote the exact couple by (A, E, i, j, k). Also define the differential of the exact couple to be $d = j \circ k$.

Notice that this actually just the data of a long exact sequence. If we look at what happens nearby $E_{p,q}$, we see that we can expand out the triangle ad infinum:

$$\cdots \longrightarrow A_{p-a,q-a'} \longrightarrow E_{p,q} \longrightarrow A_{p+a,q+a'} \longrightarrow A_{p+a+b,q+a'+b'} \longrightarrow \cdots$$

Definition 5.4.2 (Derived Couple)
Let $\mathcal{A}$ be an abelian category. Let (A, E, i, j, k) be an exact couple. Define the derived couple (A', E', i', j', k') of the exact couple by the following data.
- Define $A' = \text{im}(i)$.
- Define $E' = \text{coker}(\text{im}(d) \to \ker(d))$.
- Let the following be the unique factorization of i through $\text{im}(i)$:

$$\begin{tikzcd} A \arrow[rr, "i"] \arrow[dr, dashed, "\exists! b"'] & & A \\ & \text{im}(i) \arrow[ur, dashed, "\exists! a"'] & \end{tikzcd}$$

  Define $i' = b \circ a$.
- Define $j' : A' \to E'$ to be the connecting homomorphism in the snake lemma of the following diagram:

$$\begin{tikzcd} & \text{im}(d) \arrow[r, "k"] \arrow[d, dashed, "\exists!"'] & A \arrow[r, "i"] \arrow[d, "j"'] & A' \arrow[r] \arrow[d, "0"] & 0 \\ 0 \arrow[r] & \ker(d) \arrow[r] & E \arrow[r, "d"] & E & \end{tikzcd}$$

  (In terms of elements this is given as follows: For $x \in A'$ suppose that $i(y) = x$ for some $y \in A$. Then define $j'(x) = [j(y)]$).
- Define $k' : E' \to A'$ to be the connecting homomorphism in the snake lemma of the following the diagram:

$$\begin{tikzcd} & E \arrow[r, "d"] \arrow[d, "0"'] & E \arrow[r] \arrow[d, "k"'] \arrow[dr, "d"] & \text{coker}(d) \arrow[r] \arrow[d, dashed, "\exists!"'] & 0 \\ 0 \arrow[r] & \ker(j) \arrow[r] & A \arrow[r, "j"'] & E & \end{tikzcd}$$

  (In terms of elements this is given as follows: $k'([x]) = k(x)$).

Lemma 5.4.3
Let $\mathcal{A}$ be an abelian group. Let (A, E, i, j, k) be an exact couple. Then (A', E', i', j', k') is an exact couple.

Definition 5.4.4 (The nth Derived Couple)
Let $\mathcal{A}$ be an abelian group. Let (A, E, i, j, k) be an exact couple. Define the nth derived couple $(A^{(n)}, E^{(n)}, i^{(n)}, j^{(n)}, k^{(n)})$ inductively to be the derived couple of the $(n-1)$th derived couple.

Proposition 5.4.5
Let $\mathcal{A}$ be an abelian group. Let $a \in \mathbb{N}$. Let $(A_{\bullet,\bullet}, E_{\bullet,\bullet}, i, j, k)$ be an exact couple in $\text{Gr}_{\mathbb{Z}^2}(\mathcal{A})$ such that $\deg(i) = (1, -1)$ and $\deg(j) = (-a, a)$ and $\deg(k) = (-1, 0)$. Then the following are true.

- $\deg(i') = (1, -1)$.
- $\deg(j') = (-a-1, a+1)$.
- $\deg(k') = (-1, 0)$.
- $\left(E^{(n)}_{\bullet,\bullet}, j^{(n)} \circ k^{(n)}\right)_{n \geq a}$ is a homological spectral sequence.

5.5 The Spectral Sequence of Filtrations

Lemma 5.5.1
Let $\mathcal{A}$ be an abelian category. Let $C_\bullet \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Let $(F_pC_\bullet)_{p \in \mathbb{N}^+}$ be an ascending filtration of $C_\bullet$. Consider the following data:

- $D_{p,q} = H_{p+q}(F_pC_\bullet)$.
- $E_{p,q} = H_{p+q}\left(\frac{F_pC_\bullet}{F_{p-1}C_\bullet}\right)$.
- The short exact sequence of chain complexes $0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to \frac{F_pC_\bullet}{F_{p-1}C_\bullet}$ induces a long exact sequence:

$$\cdots \longrightarrow H_n(F_{p-1}C_\bullet) \xrightarrow{i} H_n(F_pC_\bullet) \xrightarrow{j} H_n\left(\frac{F_pC_\bullet}{F_{p-1}C_\bullet}\right) \xrightarrow{k} H_{n-1}(F_{p-1}C_\bullet) \longrightarrow \cdots$$

Then $(D_{\bullet,\bullet}, E_{\bullet,\bullet}, i, j, k)$ is an exact couple.

Definition 5.5.2 (Spectral Sequence of a Filtered Chain Complex)
Let $\mathcal{A}$ be an abelian category. Let $C_\bullet \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Let $(F_pC_\bullet)_{p \in \mathbb{N}^+}$ be an ascending filtration of $C_\bullet$. Define the spectral sequence of the filtered chain complex as follows.

- Define $E^0_{p,q} = \left(\frac{F_pC_{p+q}}{F_{p-1}C_{p+q}}\right)$.
- For $n \in \mathbb{N}^+$, define $E^n_{\bullet,\bullet}$ to be the spectral sequence associated to the exact couple $(D_{\bullet,\bullet}, E_{\bullet,\bullet}, i, j, k)$ given above.

There is a more explicit way of constructing the associated spectral sequence.

Definition 5.5.3 (Spectral Sequence Arising from Filtered Chain Complexes) Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, d_\bullet) \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Let F_pC be a filtered chain complex that is exhaustive. Define the following objects and subobjects in $\mathcal{A}$.

- Define an object $E^0_{p,q} = \frac{F_pC_{p+q}}{F_{p-1}C_{p+q}}$ and a chain complex $E^0_p = \frac{F_pC}{F_{p-1}C}$.
- For each $p, r \in \mathbb{N}$, define
$$A^r_p = \{c \in F_pC \mid d(c) \in F_{p-r}C\}$$
called approximately cycles
- Write $\eta_p : F_pC \to \frac{F_pC}{F_{p-1}C} = E^0_p$ for the sujection, which is a chain map.
- For each $p, r \in \mathbb{N}$, define
$$Z^r_p = \eta_p(A^r_p) \subseteq E^0_p$$
- For each $p, r \in \mathbb{N}$, define
$$B^{r+1}_{p-r} = \eta_{p-r}(d(A^r_p)) \subseteq E^0_{p-r}$$
- For each $p \in \mathbb{N}$, define
$$Z^\infty_p = \bigcap_{r=1}^{\infty} Z^r_p \quad \text{and} \quad B^\infty_p = \bigcup_{r=1}^{\infty} B^r_p$$

Evidently, there is a tower of subobjects of E_p^0 given by

$$0 = B_p^0 \subseteq B_p^1 \subseteq \cdots B_p^\infty \subseteq Z_p^\infty \subseteq \cdots Z_p^1 \subseteq Z_p^0 = E_p^0$$

Thus we finally define

$$E_p^r = \frac{Z_p^r}{B_p^r} \quad \text{for all } r \in R \quad \text{and} \quad E_p^\infty = \frac{Z_p^\infty}{B_p^\infty}$$

together with $d : E_p^r \to E_{p-r}^r$ to be the differential induced by the differential of C.

Lemma 5.5.4 Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, d_\bullet) \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Let F_pC be a filtered chain complex that is exhaustive. Then the following are true.

- For any $p, r \in \mathbb{N}$, $A_p^r + F_{p-1}C = A_{p-1}^{r-1}$
- For any $p, r \in \mathbb{N}$, $Z_p^r \cong A_p^r / A_{p-1}^{r-1}$
- For any $p, r \in \mathbb{N}$, there are isomorphisms

$$E_p^r = \frac{Z_p^r}{B_p^r} \cong \frac{A_p^r + F_{p-1}C}{d(A_{p+r-1}^{r-1}) + F_{p-1}C} \cong \frac{A_p^r}{d(A_{p+r-1}^{r-1}) + A_{p-1}^{r-1}}$$

Theorem 5.5.5 Let $\mathcal{A}$ be an abelian category. Let $(C_\bullet, d_\bullet) \in \mathbf{Ch}(\mathcal{A})$ be a chain complex. Let F_pC be a filtered chain complex that is exhaustive. Then the map $d : E_p^r \to E_{p-r}^r$ determines an isomorphism

$$\frac{Z_p^r}{Z_p^{r+1}} \cong \frac{B_{p-r}^{r+1}}{B_{p-r}^r}$$

Moreover, this concludes that $(E_{p,q}^r, d)$ is a spectral sequence.

5.6 The Spectral Sequence of Double Complexes

5.7 Hyperhomology and Hypercohomology

Definition 5.7.1 (Cartan-Eilenberg Resolution) Let $\mathcal{A}$ be an abelian category with enough projectives. Let $A_\bullet \in \mathbf{Ch}(\mathcal{A})$ be a chain complex as follows:

$$\cdots \longrightarrow A_{p-1} \longrightarrow A_p \longrightarrow A_{p+1} \longrightarrow \cdots$$

A Cartan-Eilenberg resolution of $A_\bullet$ is an upper half double complex $P_{\bullet,\bullet}$ together with augmentation maps $\varepsilon : P_{P,\bullet} \to A_p$ as follows:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots & P_{p-1,1} \arrow[l] \arrow[d] & P_{p,1} \arrow[l] \arrow[d] & P_{p+1,1} \arrow[l] \arrow[d] & \cdots \arrow[l] \\
\cdots & P_{p-1,0} \arrow[l] \arrow[d, "\varepsilon"] & P_{p,0} \arrow[l] \arrow[d, "\varepsilon"] & P_{p+1,0} \arrow[l] \arrow[d, "\varepsilon"] & \cdots \arrow[l] \\
\cdots & A_{p-1} \arrow[l] & A_p \arrow[l] & A_{p+1} \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

such that the following are true.

- If $A_p = 0$ then $P_{p,\bullet} = 0$
- For each $p \in \mathbb{Z}$, the chain complex $(B_\bullet, d^v)$ of boundaries where $B_q = d^h(P_{p+1,q})$ form a projective resolution of A_p. This means that the vertical chain complexes

$$\begin{tikzcd}
\vdots \arrow[d] \\
d^h(P_{p,0}) \subseteq P_{p,1} \arrow[d] \\
d^h(P_{p+1,0}) \subseteq P_{p,0} \arrow[d, "\varepsilon"] \\
A_p
\end{tikzcd}$$

are projective resolutions.

- For each $p \in \mathbb{Z}$, the chain complex $(H_\bullet, d^v)$ of the homology groups where

$$H_q = H_p(P_{\bullet,q}, d^h)$$

form a projective resolution of $H_p(A_\bullet)$. This means that the vertical chain complexes

$$\begin{tikzcd}
\vdots \arrow[d] \\
H_p(P_{\bullet,1}, d^h) \arrow[d, "d^v"] \\
H_p(P_{\bullet,0}, d^h) \arrow[d, "\varepsilon"] \\
H_p(A_\bullet)
\end{tikzcd}$$

are projective resolutions.

Theorem 5.7.2 Let $\mathcal{A}$ be an abelian category. Then every chain complex $A_\bullet \in \mathbf{Ch}(\mathcal{A})$ has a Cartan-Eilenberg resolution $P_{\bullet,\bullet} \to A_\bullet$.

Definition 5.7.3 (Left Hyperderived Functors) Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor. Define the left hyperderived functors

$$\mathbb{L}_i F : \mathbf{Ch}(\mathcal{A}) \to \mathcal{B}$$

by the following.

- For $A_\bullet$ a chain complex in $\mathcal{A}$, choose a Cartan-Eilenberg resolution $P_{\bullet,\bullet} \to A_\bullet$ of $A_\bullet$. Then

$$\mathbb{L}_i F(A_\bullet) = H_i(\text{Tot}^\oplus(F(P_{\bullet,\bullet})))$$

- For $f : A_\bullet \to B_\bullet$ a chain map, choose Cartan-Eilenberg resolutions for both complexes and consider the induced map $\overline{f} : P_{\bullet,\bullet} \to Q_{\bullet,\bullet}$. Define

$$\mathbb{L}_i F(f) = H_i(\text{Tot}^\oplus(\overline{f})) : \mathbb{L}_i F(A_\bullet) \to \mathbb{L}_i F(B_\bullet)$$

Definition 5.7.4 (Right Hyperderived Functors) Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Define the right hyperderived functors

$$\mathbb{R}^i F : \mathbf{CCh}(\mathcal{A}) \to \mathcal{B}$$

by the following.

- For $A_\bullet$ a cochain complex in $\mathcal{A}$, choose a Cartan-Eilenberg resolution $A_\bullet \to I_{\bullet,\bullet}$ of $A_\bullet$. Then

$$\mathbb{R}^i F(A_\bullet) = H^i(\text{Tot}^\oplus(F(I_{\bullet,\bullet})))$$

- For $f : A_\bullet \to B_\bullet$ a chain map, choose Cartan-Eilenberg resolutions for both complexes and consider the induced map $\overline{f} : I_{\bullet,\bullet} \to J_{\bullet,\bullet}$. Define

$$\mathbb{R}^i F(f) = H^i(\text{Tot}^\oplus(\overline{f})) : \mathbb{R}^i F(A_\bullet) \to \mathbb{R}^i F(B_\bullet)$$

Lemma 5.7.5 Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $X \in \mathcal{A}$ be an object. Then the following are true.

- Let $F : \mathcal{A} \to \mathcal{B}$ be right exact. By considering X as a chain complex with only non-zero term in degree 0, the hyperderived left functor $\mathbb{L}_i F(X) = L_i F(X)$ is the same as the usual left derived functor.
- Let $F : \mathcal{A} \to \mathcal{B}$ be left exact. By considering X as a cochain complex with only non-zero term in degree 0, the hyperderived right functor $\mathbb{R}^i F(X) = R^i F(X)$ is the same as the usual right derived functor.

Lemma 5.7.6 Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Then the following are true.

- Let $F : \mathcal{A} \to \mathcal{B}$ be right exact. Then the restriction of $\mathbb{L}_i F : \mathbf{Ch}(\mathcal{A}) \to \mathcal{B}$ to $\mathbf{Ch}^+(\mathcal{A})$ are precisely the left derived functor $L_i(H_0 F) : \mathbf{Ch}^+(\mathcal{A}) \to \mathcal{B}$ of the functor $H_0 F : \mathbf{Ch}(\mathcal{A})^+ \to \mathcal{B}$.
- Let $F : \mathcal{A} \to \mathcal{B}$ be left exact. Then the restriction of $\mathbb{R}^i F : \mathbf{CCh}(\mathcal{A}) \to \mathcal{B}$ to $\mathbf{CCh}^+(\mathcal{A})$ are precisely the left derived functor $R^i(H^0 F) : \mathbf{CCh}^+(\mathcal{A}) \to \mathcal{B}$ of the functor $H^0 F : \mathbf{CCh}(\mathcal{A})^+ \to \mathcal{B}$.

Lemma 5.7.7 Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let the following

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of chain complexes in $\mathbf{Ch}^+(\mathcal{A})$. Let $F : \mathcal{A} \to \mathcal{B}$ be right exact. Then there exists a long exact sequence

$$\cdots \longrightarrow \mathbb{L}_{i+1} F(C) \xrightarrow{\delta} \mathbb{L}_i F(A) \longrightarrow \mathbb{L}_i F(B) \longrightarrow \mathbb{L}_i F(C) \xrightarrow{\delta} \mathbb{L}_{i-1} F(A) \longrightarrow \cdots$$

Proposition 5.7.8 There is always a convergent spectral sequence

$$E^2_{pq} = (L_p F)(H_q(A)) \Rightarrow \mathbb{L}_{p+q} F(A)$$

Corollary 5.7.9 Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor. Then the following are true.

- If A is exact, then $\mathbb{L}_i F(A) = 0$ for all i.
- If $f : A \to B$ is a quasi-isomorphism, then it induces isomorphisms $\mathbb{L}_i F(A) \cong \mathbb{L}_i F(B)$

Next: dual versions of the above propositions.

6 Triangulated Categories

6.1 Axioms of a Triangulated Category

Definition 6.1.1 (Triangles)
Let $\mathcal{C}$ be a category and $T : \mathcal{C} \to \mathcal{C}$ an automorphism functor. Let $A, B, C \in \mathcal{C}$. A triangle on (A, B, C) is a triple (u, v, w) of morphisms in $\mathcal{C}$ where $u : A \to B$, $v : B \to C$, $w : C \to T(A)$.

Definition 6.1.2 (Morphisms of Triangles)
Let $\mathcal{C}$ be a category and $T : \mathcal{C} \to \mathcal{C}$ an automorphism functor. Let (u, v, w) and (u', v', w') be triangles in $\mathcal{C}$. A morphism of triangles is a triple (f, g, h) such that the following diagram commutes:

$$\begin{tikzcd}
A \arrow[r, "u"] \arrow[d, "f"'] & B \arrow[r, "v"] \arrow[d, "g"'] & C \arrow[r, "w"] \arrow[d, "h"'] & T(A) \arrow[d, "T(f)"] \\
A' \arrow[r, "u'"'] & B' \arrow[r, "v'"'] & C' \arrow[r, "w'"'] & T(A')
\end{tikzcd}$$

Definition 6.1.3 (Triangulated Categories) Let $\mathcal{C}$ be an additive category. We say that $\mathcal{C}$ is a triangulated category if there is a functor $T : \mathcal{C} \to \mathcal{C}$ and a family $\{(u, v, w) \mid u, v, w \in \text{Mor}(\mathcal{C})\}$ of triangles called exact triangles such that the following hold.

- For any morphism $u : A \to B$, there exists an exact triangle (u, v, w):

$$\begin{tikzcd}
A \arrow[r, "u"] & B \arrow[r, "v"] & C \arrow[r, dashed, "\exists w"] & T(A)
\end{tikzcd}$$

  If (u, v, w) is a triangle on (A, B, C) isomorphic to an exact triangle (u', v', w') on (A', B', C'), then (u, v, w) is also exact:

$$\begin{tikzcd}
A \arrow[r, "u"] \arrow[d, "\cong"'] & B \arrow[r, "v"] \arrow[d, "\cong"'] & C \arrow[r, "w"] \arrow[d, "\cong"'] & T(A) \arrow[d, "\cong"] \\
A' \arrow[r, "u'"'] & B' \arrow[r, "v'"'] & C' \arrow[r, "w'"'] & T(A')
\end{tikzcd}$$

  Finally, $(\text{id}_A, 0, 0)$ is exact:

$$\begin{tikzcd}
A \arrow[r, "\text{id}_A"] & A \arrow[r] & 0 \arrow[r] & T(A)
\end{tikzcd}$$

- Rotations: If (u, v, w) is an exact triangle on (A, B, C), then both rotations $(v, w, -T(u))$ and $(-T^{-1}(w), u, v)$ are exact triangles on $(B, C, T(A))$ and $(T^{-1}(C), A, B)$ respectively.
- Morphisms: Let the following be exact triangles:

$$\begin{tikzcd}
A \arrow[r, "u"] & B \arrow[r, "v"] & C \arrow[r, "w"] & T(A)
\end{tikzcd}$$

$$\begin{tikzcd}
A' \arrow[r, "u'"] & B' \arrow[r, "v'"] & C' \arrow[r, "w'"] & T(A')
\end{tikzcd}$$

  Suppose that there exists morphisms $f : A \to A'$ and $g : B \to B'$ such that $g \circ u = u' \circ f$. Then there exists $h : C \to C'$ such that (f, g, h) is a morphism of triangles:

$$\begin{tikzcd}
A \arrow[r, "u"] \arrow[d, "f"'] & B \arrow[r, "v"] \arrow[d, "g"'] & C \arrow[r, "w"] \arrow[d, dashed, "\exists h"'] & T(A) \arrow[d, "T(f)"] \\
A' \arrow[r, "u'"'] & B' \arrow[r, "v'"'] & C' \arrow[r, "w'"'] & T(A')
\end{tikzcd}$$

- The Octahedral Axiom: Let the following be exact triangles:

$$A \xrightarrow{u} B \xrightarrow{j} C' \xrightarrow{k} T(A)$$

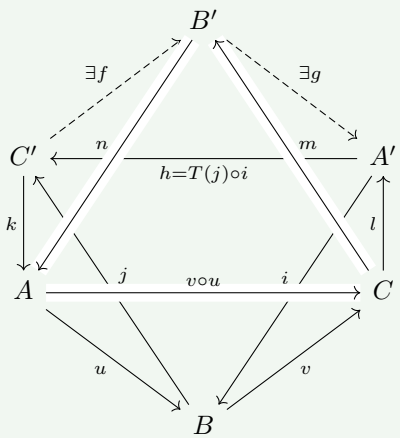
$$B \xrightarrow{v} C \xrightarrow{l} A' \xrightarrow{i} T(B)$$

$$A \xrightarrow{v \circ u} C \xrightarrow{m} B' \xrightarrow{n} T(A)$$

Then there exists an exact triangle:

$$C' \xrightarrow{f} B' \xrightarrow{g} A' \xrightarrow{h} T(C')$$

such that $l = g \circ m$, $k = n \circ f$, $h = T(j) \circ i$, $i \circ g = T(u) \circ n$ and $f \circ j = m \circ v$. In other words, the following diagram commutes:



Where we abused notation by drawing $k : C' \to T(A)$ as a morphism $C' \to A$ etc so that the drawing becomes compact.

Lemma 6.1.4 Let $(\mathcal{C}, T)$ be a triangulated category. Let (u, v, w) be an exact triangle. Then $v \circ u$, $w \circ v$ and $T(u) \circ w$ are 0 in $\mathcal{C}$.

Lemma 6.1.5 Let $(\mathcal{C}, T)$ be a triangulated category. Let (f, g, h) be a morphism of exact triangles. If both f and g are isomorphisms, then h is an isomorphism.

Definition 6.1.6 (Distinguished Triangles in K($\mathcal{A}$)) Let $\mathcal{A}$ be an abelian category. We say that a triangle

$$A \longrightarrow B \longrightarrow C \longrightarrow T(A)$$

is distinguished in $K(\mathcal{A})$ if it is isomorphic to a triangle of the form

$$X \xrightarrow{f} Y \longrightarrow C(f) \longrightarrow T(X)$$

Lemma 6.1.7 Let $\mathcal{A}$ be an abelian category. Then $K(\mathcal{A})$, $K^+(\mathcal{A})$, $K^-(\mathcal{A})$ and $K^b(\mathcal{A})$ are all triangulated categories with distinguished triangles given by the above definition.

6.2 Morphisms of Triangulated Categories

Definition 6.2.1 (Morphisms of Triangulated Categories) Let $\mathcal{C}$ and $\mathcal{D}$ be triangulated categories. A morphism from $\mathcal{C}$ to $\mathcal{D}$ is a functor $F : \mathcal{C} \to \mathcal{D}$ such that the following are true.

- F is an additive functor
- F commutes with the translation functor. If T is the automorphism of $\mathcal{C}$ and S is the automorphism of $\mathcal{D}$, then
$$F \circ T = S \circ F$$
- F sends exact triangles to exact triangles

7 Derived Categories

7.1 Localization of Categories

Definition 7.1.1 (Localization of a Category) Let $\mathcal{C}$ be a category and let S be a collection of morphisms in $\mathcal{C}$. A localization of $\mathcal{C}$ with respect to S is a category $S^{-1}\mathcal{C}$ together with a functor $q : \mathcal{C} \to S^{-1}\mathcal{C}$ such that the following are true.

- For all $s \in S$, $q(s)$ is an isomorphism in $S^{-1}\mathcal{C}$
- If $F : \mathcal{C} \to \mathcal{D}$ is a functor such that $F(s)$ is an isomorphism for all $s \in S^{-1}\mathcal{C}$, then there exists a unique functor $G : S^{-1}\mathcal{C} \to \mathcal{D}$ such that the following diagram commute:

$$\begin{tikzcd} \mathcal{C} \arrow[r, "q"] \arrow[dr, "F"'] & S^{-1}\mathcal{C} \arrow[d, dashed, "\exists! G"] \\ & \mathcal{D} \end{tikzcd}$$

Lemma 7.1.2 Let $\mathcal{A}$ be an abelian category. Then $K(\mathcal{A})$ is a localization of $\mathcal{A}$ with respect to all homotopy equivalences.

Not all localizations are well defined by set-theoretic issues. Morphisms that one wants to invert may not form a set or even a collection. We will give a way of explicitly constructing the localization of some specific categories below.

Definition 7.1.3 (Multiplicative System)

Definition 7.1.4 (Locally Small Multiplicative System)

Theorem 7.1.5 (Gabriel-Zisman Theorem)

Corollary 7.1.6 Let $\mathcal{C}$ be a category containing the zero object 0 and let $q : \mathcal{C} \to S^{-1}\mathcal{C}$ be a localization of $\mathcal{C}$. Then $q(X) \cong 0$ if and only if the S contains the 0 map $0 : X \to X$.

Corollary 7.1.7 Let $\mathcal{C}$ be a category and let $q : \mathcal{C} \to S^{-1}\mathcal{C}$ be a localization of $\mathcal{C}$. If $\mathcal{C}$ is additive, then $S^{-1}\mathcal{C}$ and q are both additive.

7.2 Derived Categories

Definition 7.2.1 (Derived Categories) Let $\mathcal{A}$ be an abelian category and let $\mathbf{CCh}(\mathcal{A})$ be any category of chain complexes of $\mathcal{A}$. Define the derived category

$$D(\mathcal{A}) = \mathbf{CCh}(\mathcal{A})[\mathcal{W}^{-1}]$$

of $\mathcal{A}$ where $\mathcal{W}$ is all the quasi-isomorphisms in $\mathbf{CCh}(\mathcal{A})$.

Respectively, for $\mathbf{CCh}^+(\mathcal{A})$ and $\mathbf{CCh}^b(\mathcal{A})$ define their derived categories to be the localization of the categories with respect to quasi-isomorphisms, denoted by $D^+(\mathcal{A})$ and $D^b(\mathcal{A})$ respectively.

Theorem 7.2.2 Let $\mathcal{A}$ be an abelian category. Let $K(\mathcal{A})$ be the homotopy category of chain complexes of $\mathcal{A}$. Let $\mathcal{W}$ be all the quasi-isomorphisms in $\mathbf{CCh}(\mathcal{A})$. Then there is an equivalence of categories

$$D(\mathcal{A}) = K(\mathcal{A})[\mathcal{W}^{-1}]$$

Definition 7.2.3 (Distinguished Triangles in D($\mathcal{A}$)) Let $\mathcal{A}$ be an abelian category. We say that a triangle

$$A \longrightarrow B \longrightarrow C \longrightarrow T(A)$$

is distinguished in $D(\mathcal{A})$ if it is isomorphic to a triangle of the form

$$X \xrightarrow{f} Y \longrightarrow C(f) \longrightarrow T(X)$$

Theorem 7.2.4 Let $\mathcal{A}$ be an abelian category. Then $D(\mathcal{A})$ is a triangulated triangles with distinguished triangles given by the above definition.

Definition 7.2.5 (Full Subcategory of Complexes) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories such that $\mathcal{A}$ is a subcategory of $\mathcal{B}$. Define

$$D_{\mathcal{A}}(\mathcal{B})$$

to be the full triangulated subcategory of complexes with cohomology in $\mathcal{A}$.

7.3 Derived Categories in Terms of Resolutions

Definition 7.3.1 (The Category of Injective / Projective Objects)
Let $\mathcal{A}$ be an abelian category.

- Define the category of injective objects $\mathbf{Inj}(\mathcal{A})$ to be the full subcategory of $\mathcal{A}$ consisting of injective objects.
- Define the category of projective objects $\mathbf{Proj}(\mathcal{A})$ to be the full subcategory of $\mathcal{A}$ consisting of projective objects.

Lemma 7.3.2
Let $\mathcal{A}$ be an abelian category. Then $\mathbf{Inj}(\mathcal{A})$ and $\mathbf{Proj}(\mathcal{A})$ are abelian categories.

Proposition 7.3.3
Let $\mathcal{A}$ be an abelian category. Then the following are true.

- If $\mathcal{A}$ has enough injectives, then the induced functor by inclusion

$$K^+(\text{Inj}(\mathcal{A})) \to D^+(\mathcal{A})$$

  is an equivalence of categories.
- If $\mathcal{A}$ has enough projectives, then the induced functor by inclusion

$$K^-(\text{Proj}(\mathcal{A})) \to D^-(\mathcal{A})$$

  is an equivalence of categories.

7.4 Derived Functors of Derived Categories

Definition 7.4.1 (Total Right Derived Functors) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : K(\mathcal{A}) \to K(\mathcal{B})$ be a morphism of triangulated categories. A total right derived functor of F is a morphism

$$RF : D(\mathcal{A}) \to D(\mathcal{B})$$

together with a natural transformation $\xi : (K(\mathcal{A}) \to K(\mathcal{B}) \to D(\mathcal{B})) \Rightarrow (K(\mathcal{A}) \to D(\mathcal{A}) \to D(\mathcal{B}))$ using the following diagram:

$$\begin{tikzcd} K(\mathcal{A}) \arrow[r, "F"] \arrow[d, "\exists!"'] & K(\mathcal{B}) \arrow[d, "\exists!"] \\ D(\mathcal{A}) \arrow[r, "RF"'] & D(\mathcal{B}) \end{tikzcd}$$

from the top-right path to the lower-left path which is universal in the following sense. If $G : D(\mathcal{A}) \to D(\mathcal{B})$ is another morphism equipped with the same natural transformation χ, then there exists a unique natural transformation

$$\eta : RF \Rightarrow G$$

such that $\chi = \eta \circ \xi$.

Definition 7.4.2 (Total Left Derived Functors) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : K(\mathcal{A}) \to K(\mathcal{B})$ be a morphism of triangulated categories. A total right derived functor of F is a morphism

$$LF : D(\mathcal{A}) \to D(\mathcal{B})$$

together with a natural transformation $\xi : (K(\mathcal{A}) \to D(\mathcal{A}) \to D(\mathcal{B})) \Rightarrow (K(\mathcal{A}) \to K(\mathcal{B}) \to D(\mathcal{B}))$ using the following diagram:

$$\begin{tikzcd} K(\mathcal{A}) \arrow[r, "F"] \arrow[d, "\exists!"'] & K(\mathcal{B}) \arrow[d, "\exists!"] \\ D(\mathcal{A}) \arrow[r, "RF"'] & D(\mathcal{B}) \end{tikzcd}$$

from the lower-left path to the top-right path which is universal in the following sense. If $G : D(\mathcal{A}) \to D(\mathcal{B})$ is another morphism equipped with the same natural transformation χ, then there exists a unique natural transformation

$$\eta : G \Rightarrow LF$$

such that $\chi = \eta \circ \xi$.

Lemma 7.4.3 Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : K(\mathcal{A}) \to K(\mathcal{B})$ be a morphism of triangulated categories. If F is exact, then F is its own left and right total derived functor.

Theorem 7.4.4 Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor. Then the following are true.

- If F is left exact and $\mathcal{A}$ has enough injectives, then

$$\mathbb{R}^i F = H^i(RF) : \mathbf{CCh}(\mathcal{A}) \to \mathcal{B}$$

- If F is right exact and $\mathcal{A}$ has enough projectives, then

$$\mathbb{L}_i F = H_i(LF) : \mathbf{Ch}(\mathcal{A}) \to \mathcal{B}$$

Notice that in particular, if $X \in \mathcal{A}$ is an object considered as a chain complex in degree 0, we have seen that $\mathbb{L}_i F = L_i F : \mathcal{A} \to \mathcal{B}$ hence $L_i F = H_i(LF) : \mathcal{A} \to \mathcal{B}$. This is similar for the dual version.